MS3.08 Introduction to Computer Controlled Experiments

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Aims

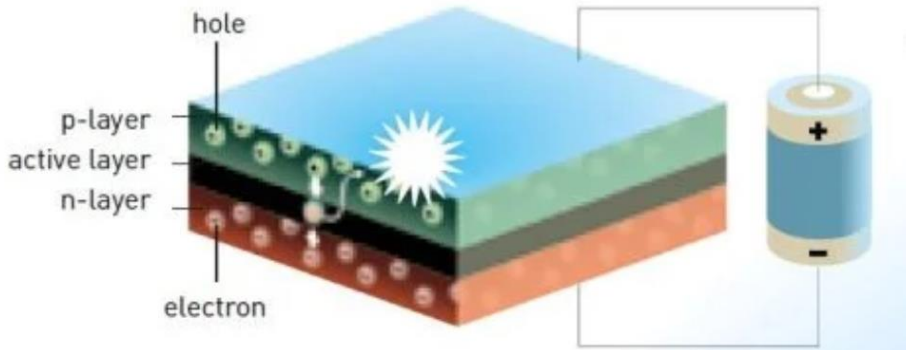
1. Characterise the **I–V** responses of **LEDs** to demonstrate the Shockley diode behaviour.

2. Investigate the relation between **turn-on voltage** of different semiconductor materials and **bandgap energy** and **emission colour** of LEDs.

3. Investigate how **temperature** influences the I–V characteristics and turn-on voltages of LEDs by comparing performances at room temperature and under liquid-nitrogen cooling.

4. Investigate how differences in turn-on voltage affect **brightness** in parallel-connected LEDs of different emission colours.

Fundamental Principles of LEDs



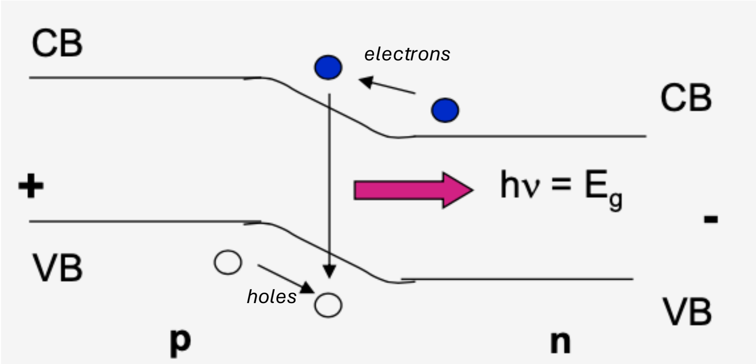
Taken from [1]

An **LED** emits light when a **forward bias** is applied across a **p–n junction** in a compound semiconductor.

- Energy barrier reduced by applying forward bias, promoting electron diffusion

LED operating principle

- **Electrons** (n-type) and **holes** (p-type) injected across p–n junction and **recombine**
- **Radiative recombination** releases energy as photons
- Photon energy (**colour**) determined by the **semiconductor bandgap** E_g
- Efficient light emission requires **direct bandgap** semiconductors



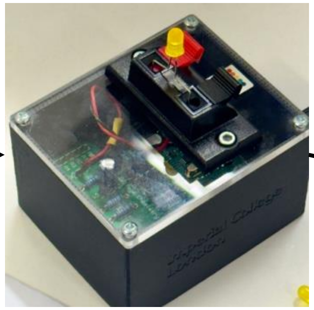
taken from EPS L6

[1] Nobel Prize Outreach AB, *The Nobel Prize in Physics 2014 – Popular Information*, https://www.nobelprize.org/prizes/physics/2014/popular-information/ (accessed 24 January 2026).

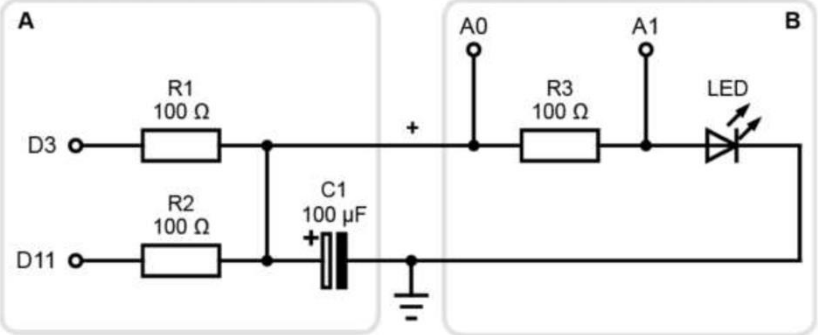
Experimental Methodology

Serial Port Configuration

Baud rate	Timeout	Serial format
9600 baud	2 s	8N1



Picture of Arduino microcontroller

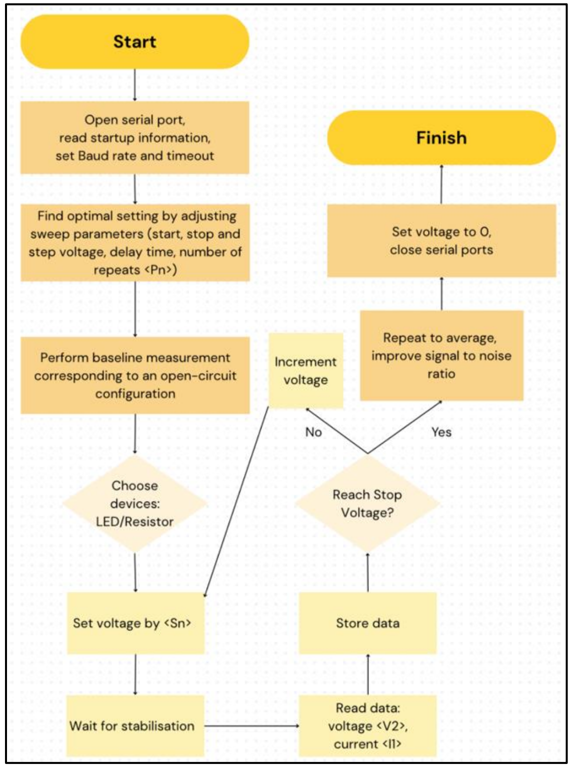


Schematic circuit setup for measurement, figure taken from experimental manual

Experimental Setup

- **LED samples**: red, yellow, green, blue, white
- **Control**: standard resistor
- **Arduino microcontroller:** voltage sourcing, current sensing, timing control, serial data acquisition
- **Measurement circuit:** series LED–resistor configuration
- **Environmental conditions:** room temperature and liquid N_2 cooling

Acquisition Workflow



Data Handling

- Real-time serial acquisition →
- Live I–V plotting (Python) →
- Turn-on voltage extraction (Origin) →
- Post-analysis & visualisation (Python)

Experiment condition optimisation 1: delay time

- **Delay time** (waiting period between setting the voltage and recording data) was varied between **0.01 and 0.30 s**.
- Short delay times resulted in **unstable resistance values** due to incomplete signal stabilisation.
- Increasing the delay time led to **convergence of the resistance**, obtained by averaging **five repeated measurements**.
- No further improvement was observed beyond **0.1 s**, which was therefore chosen as the **minimum stable delay** balancing accuracy and efficiency.

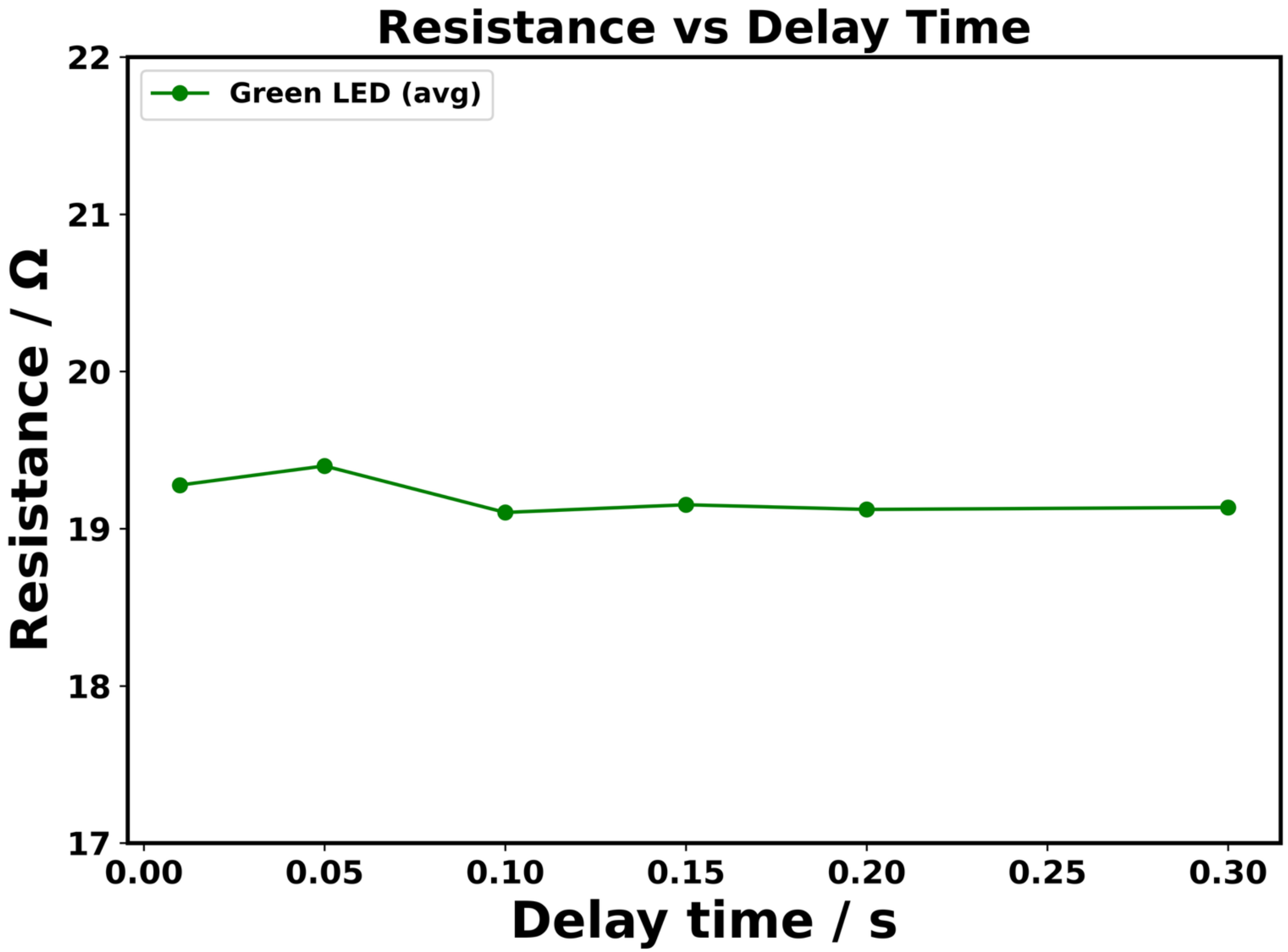
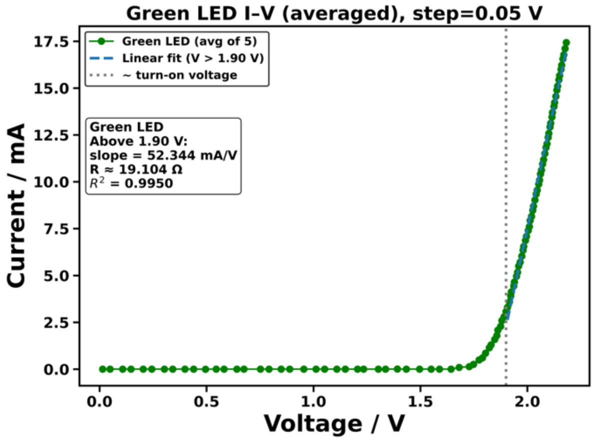


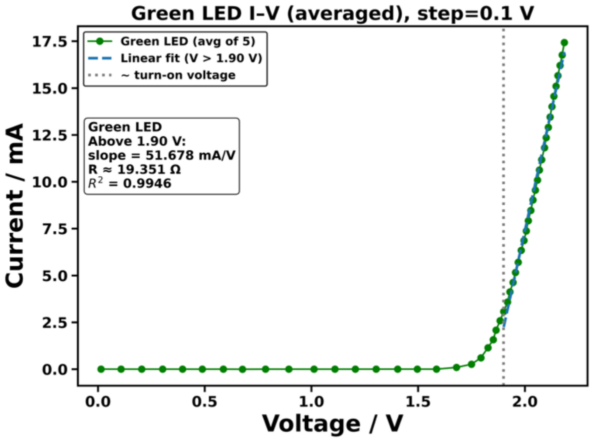
Figure for separate runs included in the SI

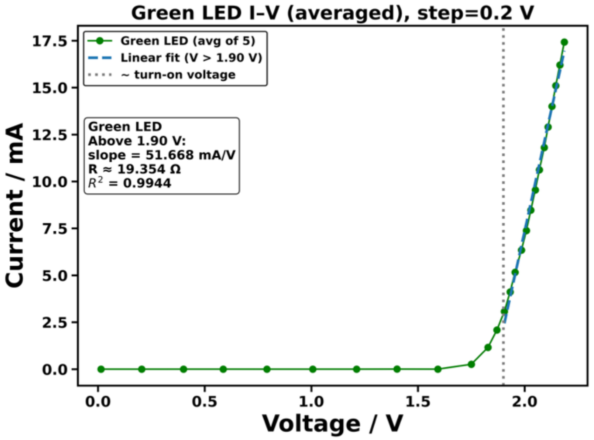
Experiment condition optimisation 2: step voltage

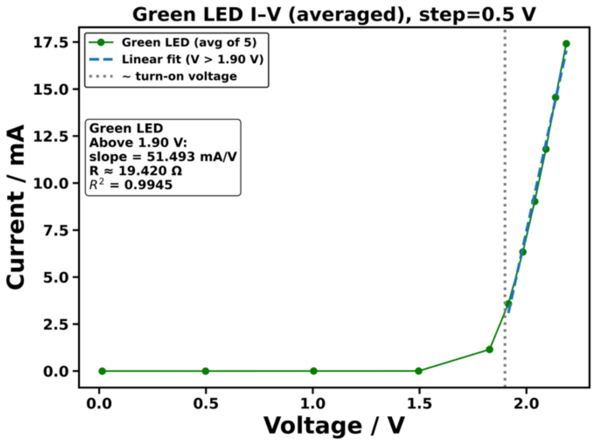
- **Step voltage** (increment in applied voltage between successive measurements) was varied from **0.05 to 1.0 V**.
- The step size controls both **data resolution** and **total acquisition time**.
- Large step sizes produce **sparse I–V data**, limiting reliable curve fitting near turn-on.
- Very small step sizes significantly **increase measurement time** without clear improvement in accuracy.
- A **step voltage of 0.1 V** was selected as the minimum step size that preserves **turn-on resolution** while maintaining **experimental efficiency**.

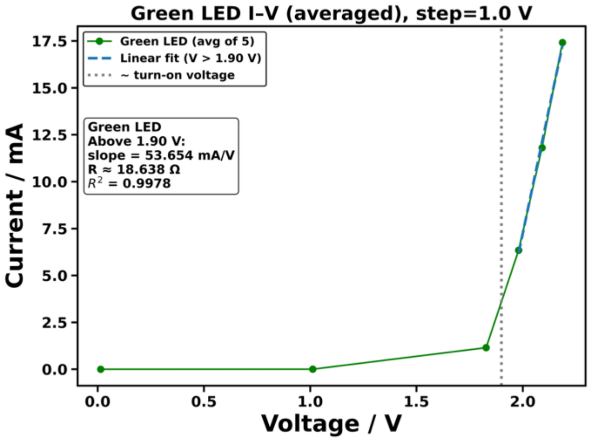
(Turn-on region resolution directly determines the reliability of downstream derivative and semi-log analyses.)











I-V Characteristics

- The **standard resistor** shows a linear I–V relationship across the full voltage range, validating the measurement setup and confirming **Ohm’s law** behaviour.
- All LEDs exhibit **non-linear diode behaviour**, with negligible current below turn-on followed by a rapid increase, consistent with **p–n junction** physics.
- The **turn-on voltage increases systematically** from red to blue/white LEDs, reflecting the increase in **semiconductor bandgap energy**.
- Above turn-on, the LED I–V curves become **approximately linear over a limited range**, allowing an **effective resistance** to be defined for comparison and fitting purposes.

*The maximum current appears to be **automatically limited by the Arduino-based measurement system**, resulting in a capped upper current across all LEDs.*

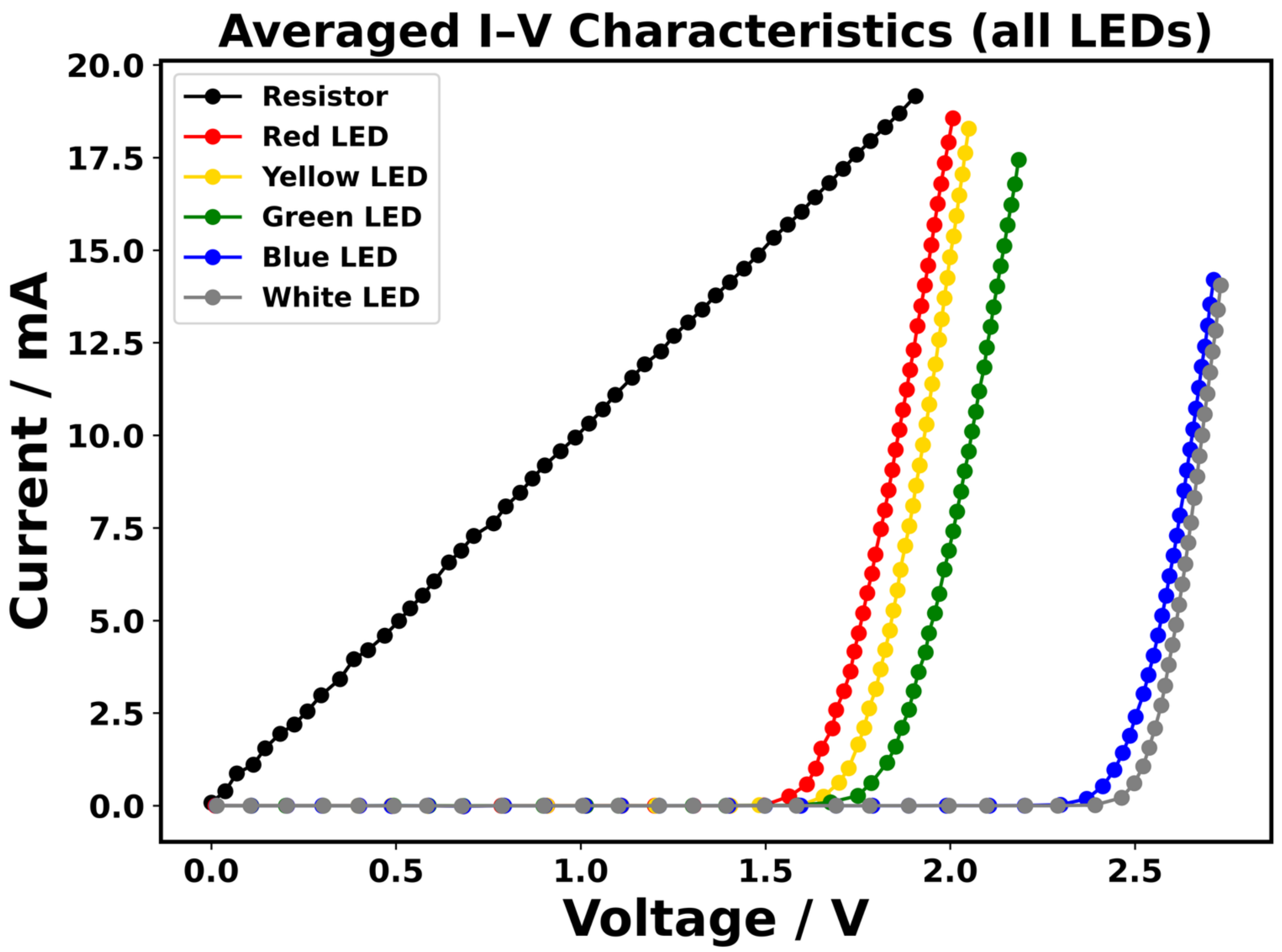
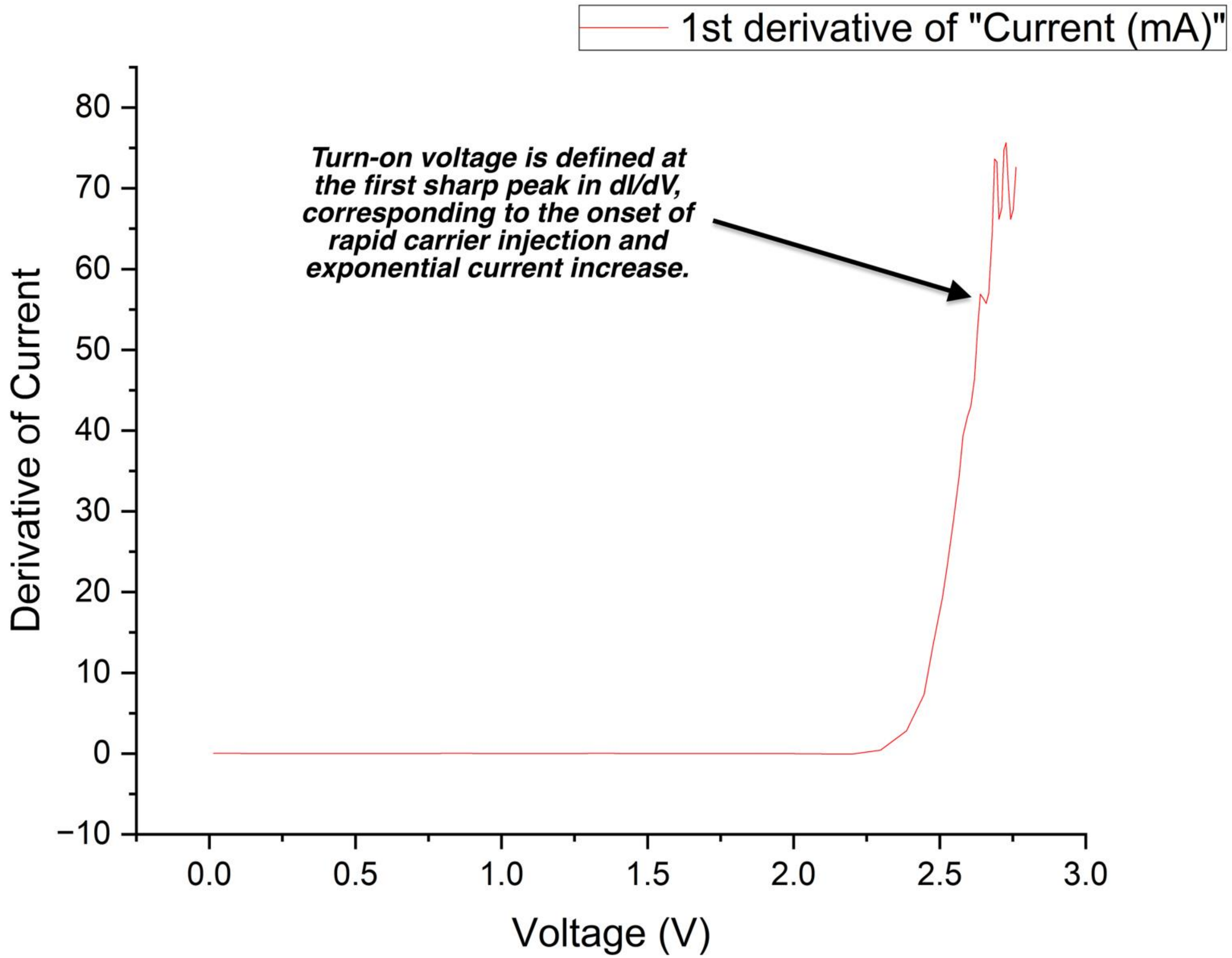


Figure for separate runs included in the SI

Finding the turn-on voltage using *Origin*

Justification

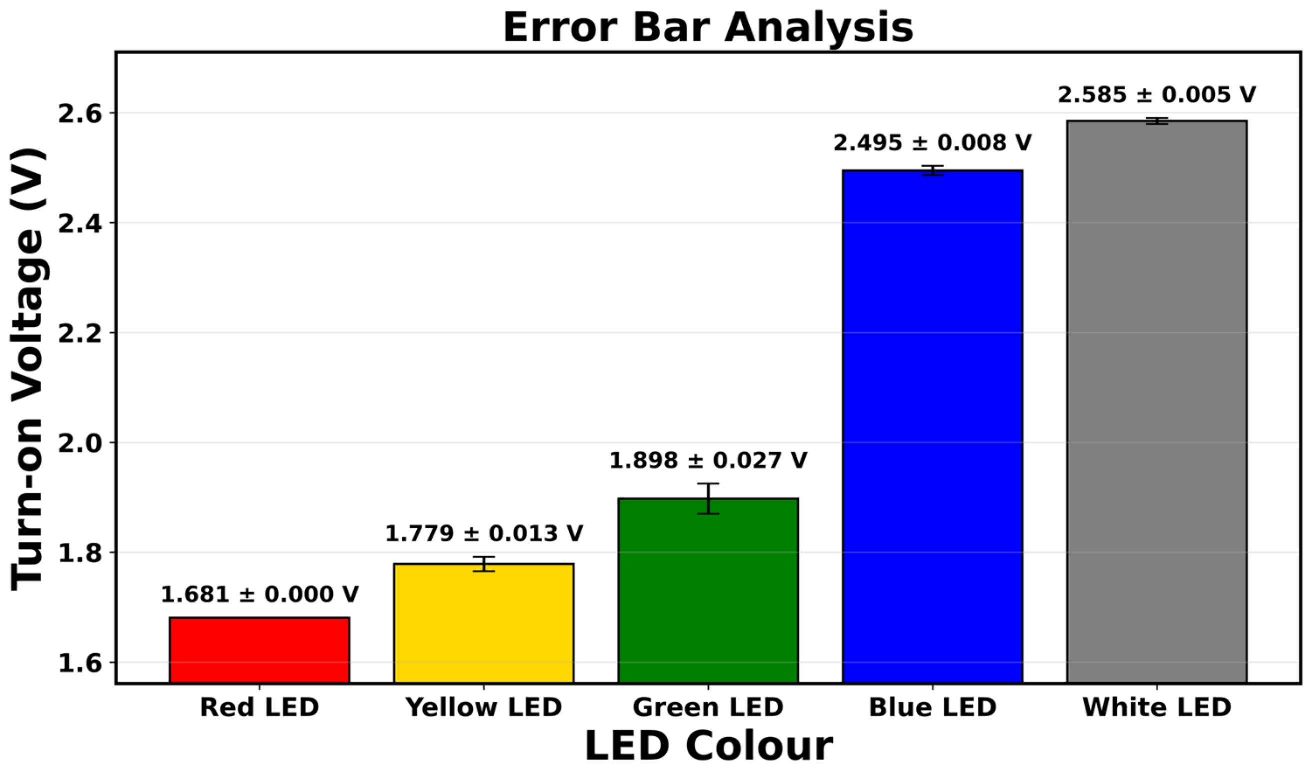
- Below turn-on, current barely increases due to **limited carrier injection**
- At turn-on, carrier injection rises rapidly and current becomes **exponential**
- This transition produces a sharp peak in the first derivative dI/dV
- The first peak in dI/dV therefore defines the turn-on voltage objectively



This example figure is exported from Origin

Turn-on Voltage Analysis

- The measured turn-on voltage increases systematically from **red → yellow → green → blue → white LEDs**.
- This trend is consistent with the **increase in semiconductor bandgap energy** for shorter-wavelength emission.
- The measured values fall within, or close to, **theoretical ranges reported in the literature**, despite the LEDs being commercially packaged devices with unspecified materials.
- The small error bars indicate **good repeatability under identical measurement conditions**, while the absolute accuracy is limited by the **instrument resolution and turn-on extraction method**.

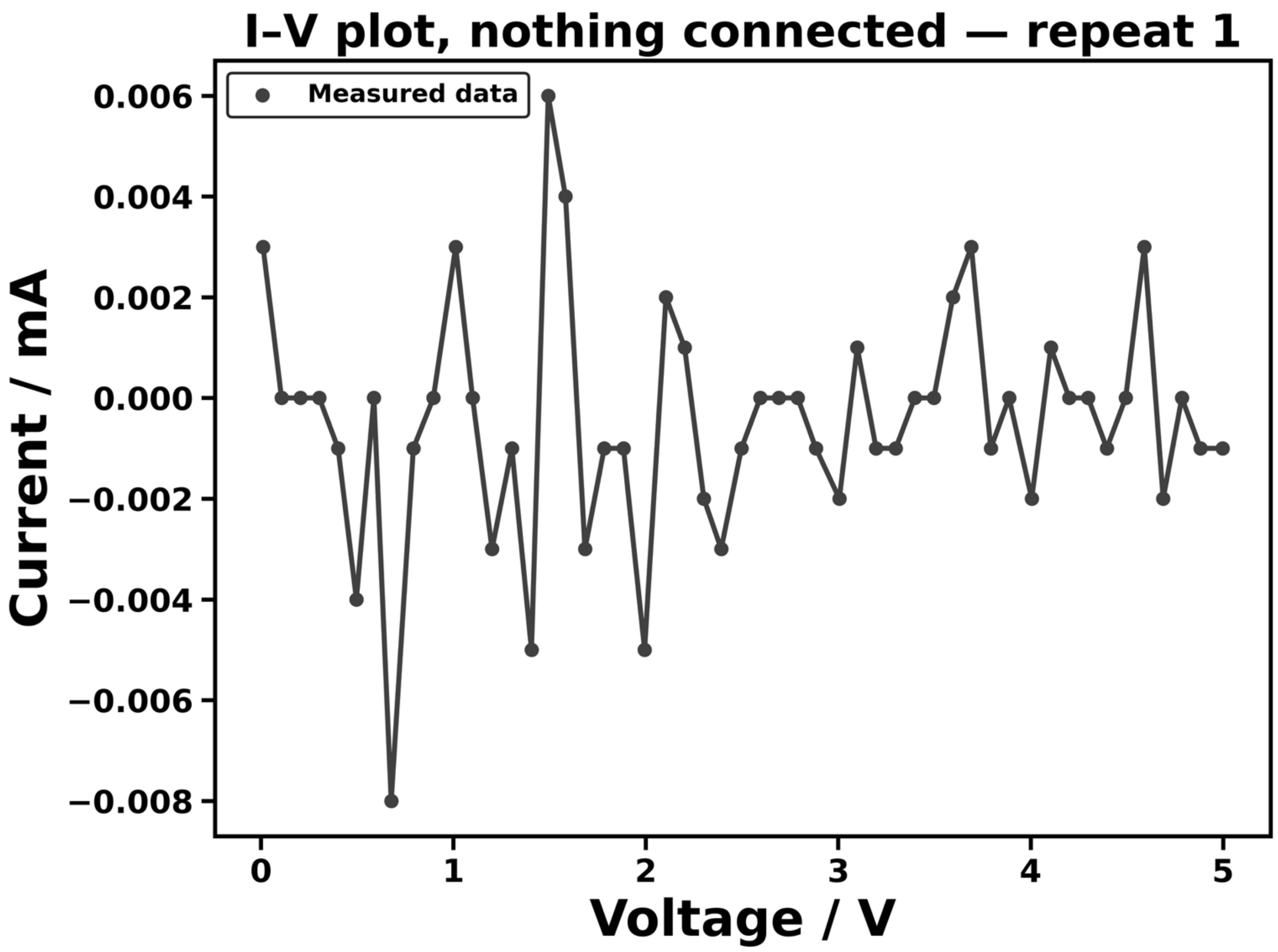


Colour	Red	Yellow	Green	Blue	White
Measured average turn-on voltage	1.681	1.779	1.898	2.495	2.585
Theoretical Turn-on voltage [2]	1.630 - 2.030	2.100 – 2.180	1.900 – 4.00	2.480 – 3.700	2.700 – 3.500

[2]. Toshiba Electronic Devices & Storage Corporation, Discrete Semiconductors – Chapter 5.3, https://toshiba.semicon-storage.com/eu/semiconductor/knowledge/e-learning/discrete/chap5/chap5-3.html (accessed 24 January 2026).

Baseline current offset as an error source

- A **baseline (open-circuit) measurement** was performed with no LED or resistor connected.
- Current calculated as $I = (A0 - A1)/R$; for an ideal open circuit, $A0 \approx A1$ and the true current is **zero**.
- Small **non-zero current offsets** were observed, arising from **ADC noise, offset errors, and finite resolution** of the measurement system.
- This baseline defines the **systematic current offset** and sets the **lower limit of measurable current**, which propagates into turn-on voltage uncertainty.



- *±μA levels of baseline noise observed*
- *Figure for all runs included in the SI*

$\log(I) - V$ plot (post turn-on voltage)

Post turn-on I–V characteristics

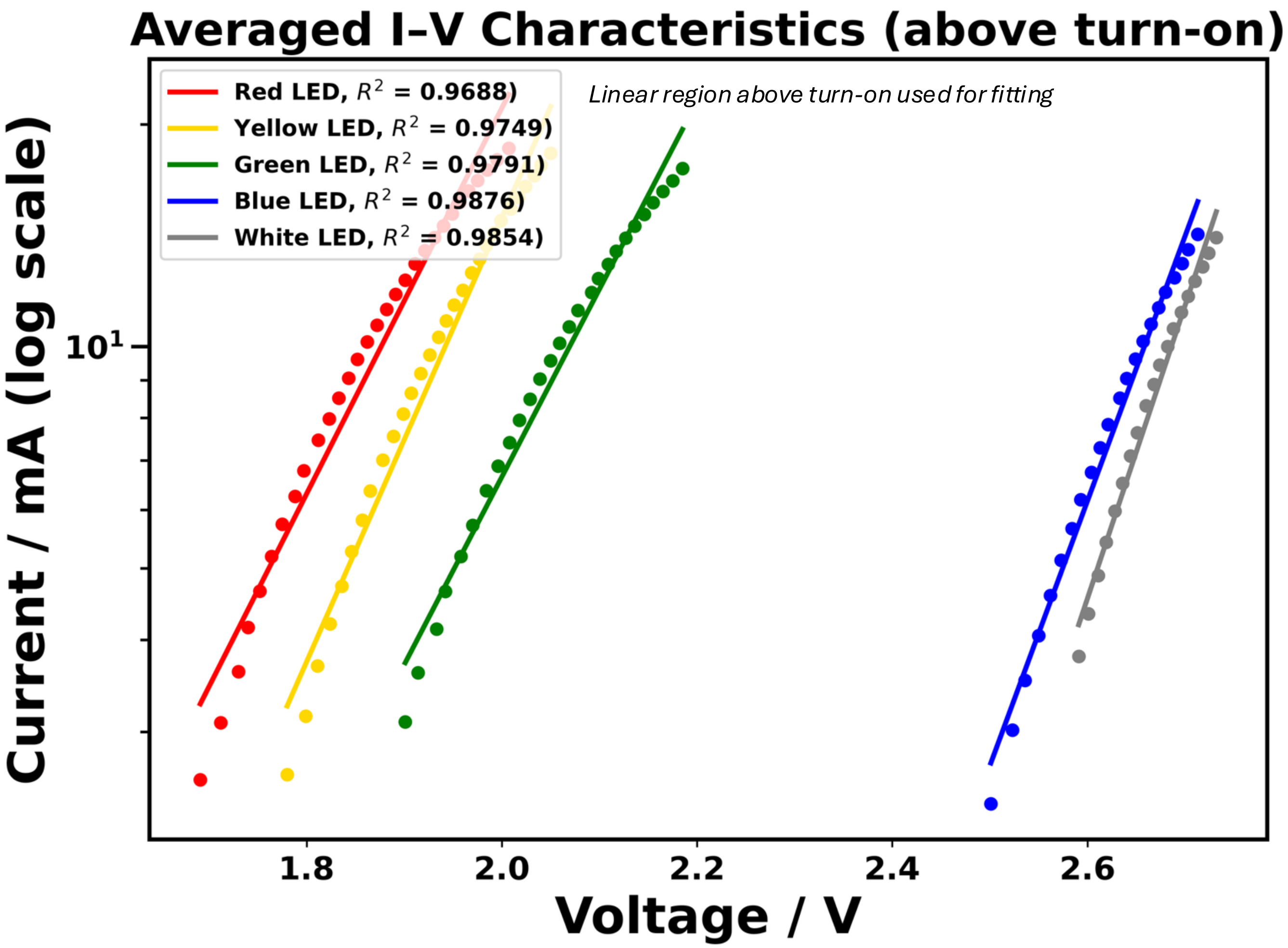
- Above turn-on voltage, LED current increases rapidly with applied voltage
- **Carrier injection** across the p–n junction dominates conduction

Semi-log linear trend

- Plotting current on **logarithmic scale** reveals an approximately linear region, indicating **consistent exponential increase** of current with voltage
- The high linearity confirms **stable diode operation** above turn-on voltage

Deviations from ideal behaviour

- Small deviations arise from **resistive losses** and **device heating**
- Different LED colours show different slopes due to **material differences**



LED colour	Red	Yellow	Green	Blue	White
Slope (dec/V)	2.61514	3.01107	2.54636	3.59099	3.97275

Theory

Shockley diode equation

$$I_D = I_S(e^{\frac{qV}{nkT}} - 1)$$

At voltages above turn-on, the exponential term dominates:

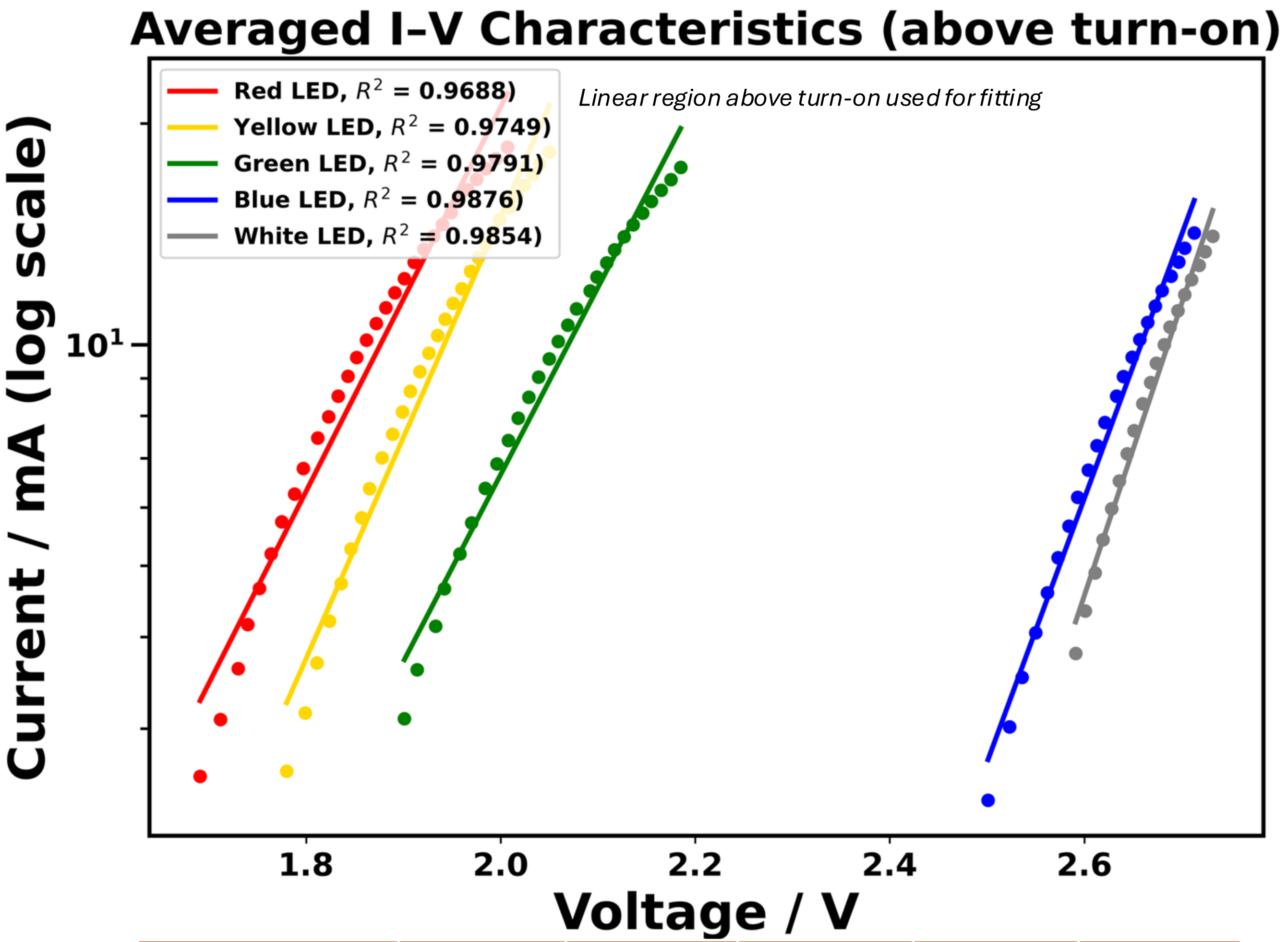
$$I_D \approx I_S e^{\frac{qV}{nkT}}$$

Taking the logarithm linearises the relationship:

$$\log I_D = \frac{q}{nkT \ln 10} \cdot V + \log I_S$$

Therefore, the slope of the $\log(I) - V$ plot is **inversely** proportional to the **effective ideality factor** n, reflecting material and non-ideal effects.

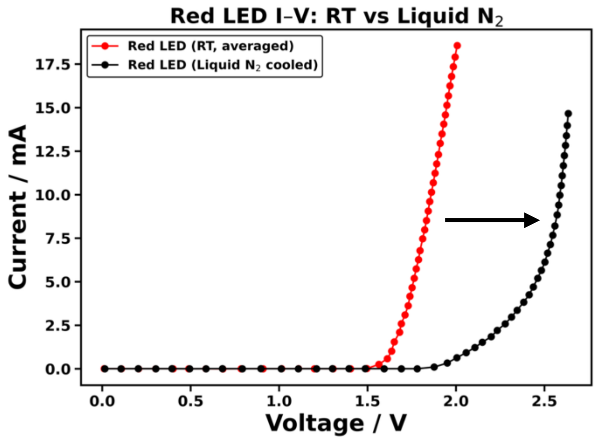
More detailed justification included in SI

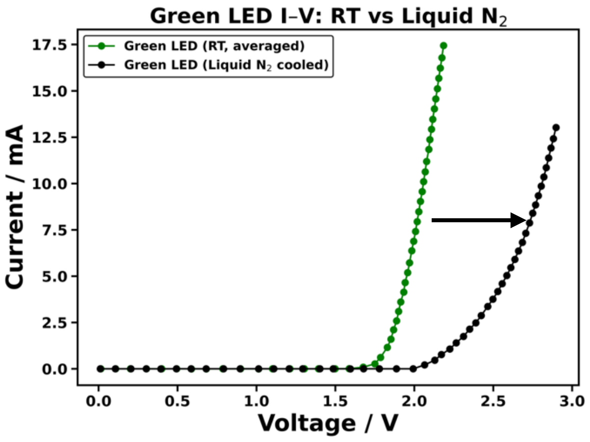


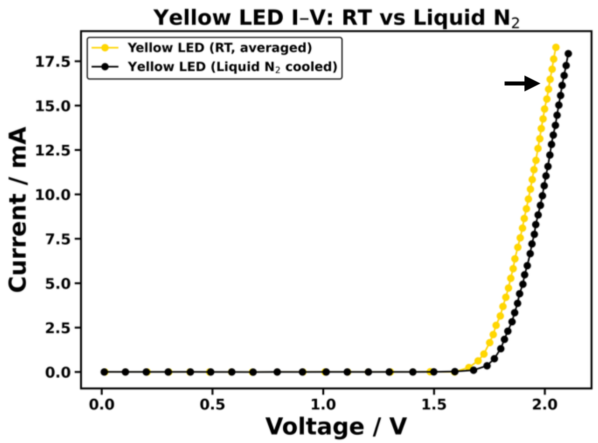
LED colour	Red	Yellow	Green	Blue	White
Slope (dec/V)	2.61514	3.01107	2.54636	3.59099	3.97275

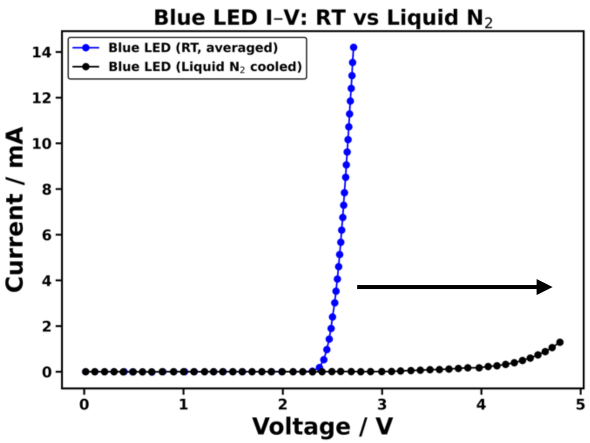
Temperature Dependence of LED Performance 1

- A **systematic increase** in turn-on voltage was observed when the LEDs were **cooled** from room temperature to liquid nitrogen temperature (77 K).
- At lower temperatures, a **higher applied voltage** is required to achieve comparable carrier injection and current levels. [3]
- Cooling reduces lattice vibrations, leading to **bandgap widening** due to **lattice contraction** in the semiconductor material [4].
- The increased bandgap raises the energy required for electron–hole recombination, resulting in a **higher turn-on voltage** (and, in principle, **shorter-wavelength emission**).



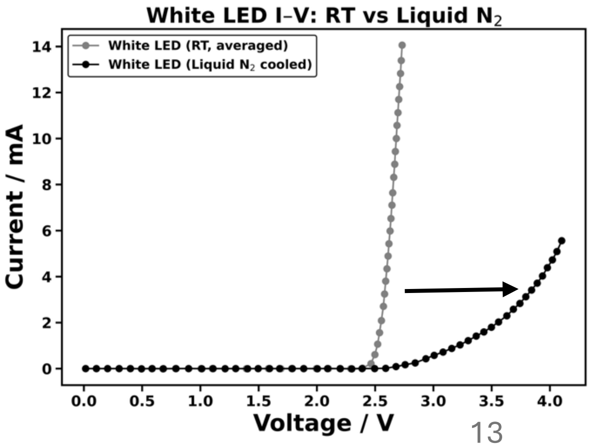






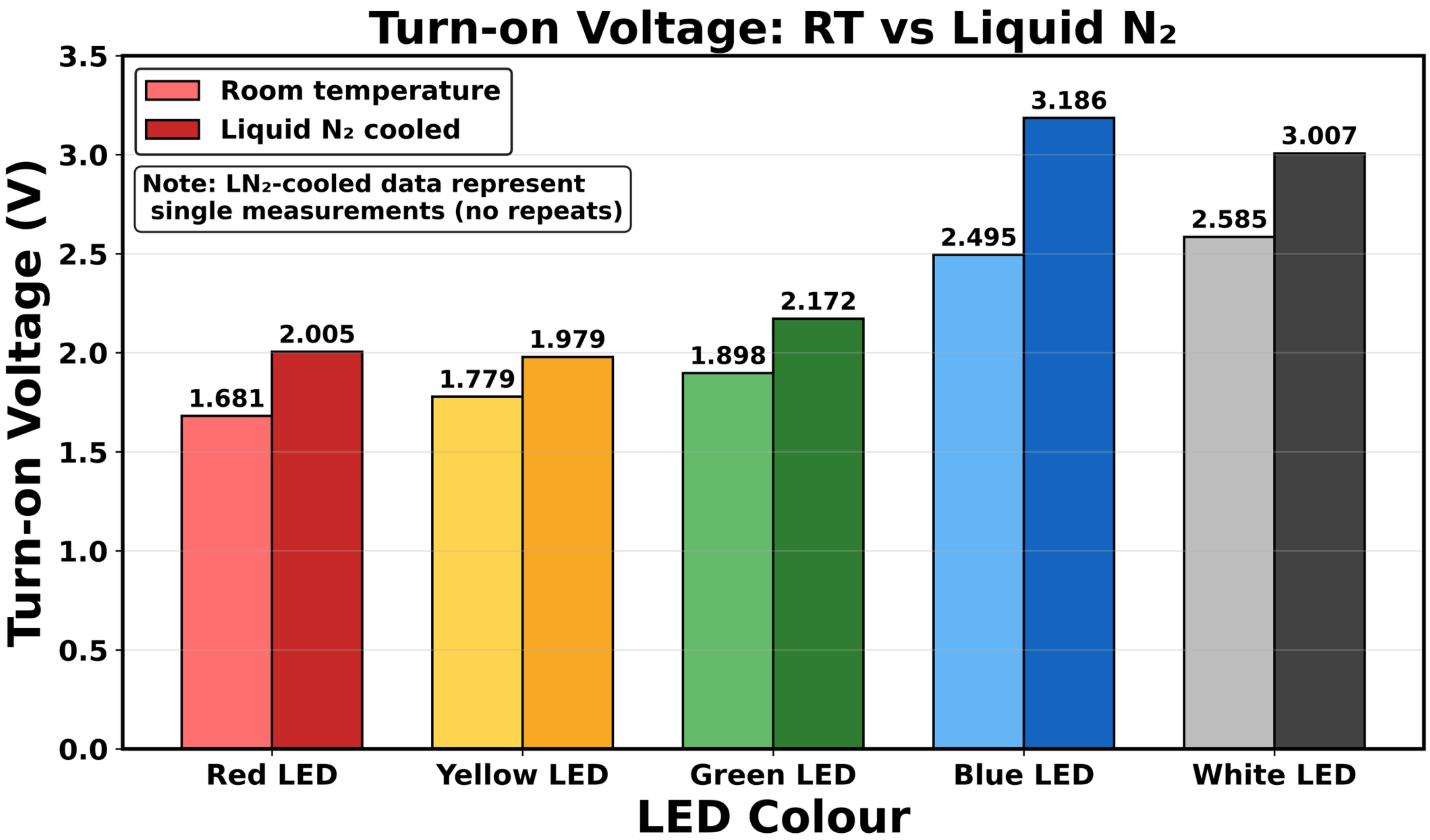
Experimental notes

- *Liquid N_2–cooled measurements were limited to a **single** sweep per LED due to laboratory constraints.*
- *Emission wavelength shift not directly measured; inferred from band-gap dependence.*



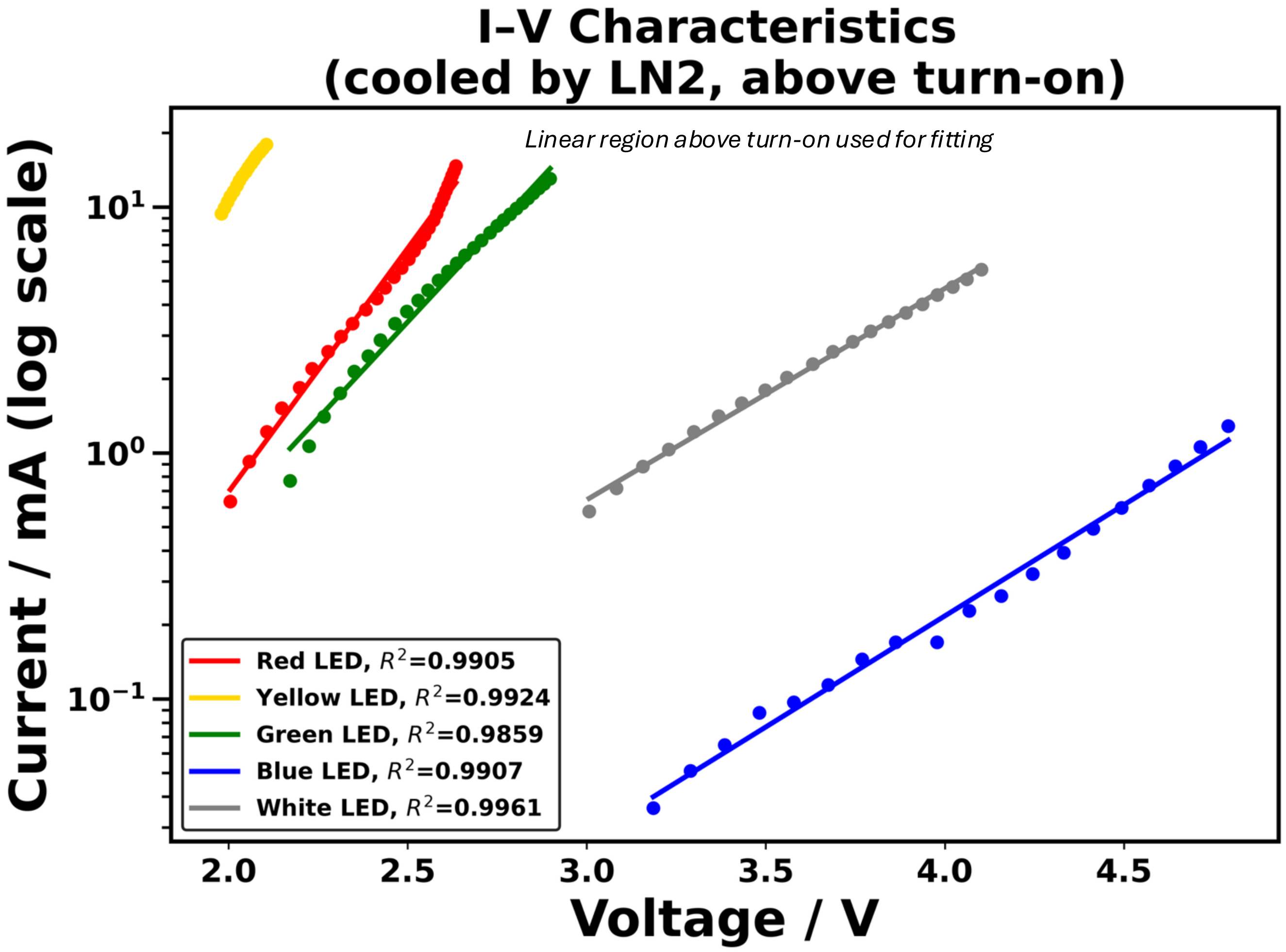
[3]. C. Park, J. Park, S. Min, J.-I. Shim and D.-S. Shin, ACS Photonics, 2019, 6, 2169–2176.
[4]. LibreTexts, Light Emitting Diodes, https://eng.libretexts.org/Bookshelves/Materials_Science/Supplemental_Modules_%28Materials_Science%29/Semiconductors/Light_Emitting_Diodes (accessed 24 January 2026).

Temperature Dependence of LED Performance 2



$\log(I) - V$ plot (post turn-on voltage) - cooled by LN2

- $\log(I) - V$ curves above turn-on are linear for all LEDs at 77 K.
- At LN_2 temperature, **reduced thermal noise suppress non-ideal transport effects**, resulting in a more well-defined $\log(I) - V$ relationship and higher R^2.
- Reduced slopes arise from **lower thermal voltage** and **higher apparent ideality factor** at low temperature.
- **Material-dependent** differences between LEDs remain observable.



LED colour	Red	Yellow	Green	Blue	White
Slope (dec/V)	1.97967	2.24007	1.56792	0.90316	0.85926

Apparent ideality factors

Extraction method

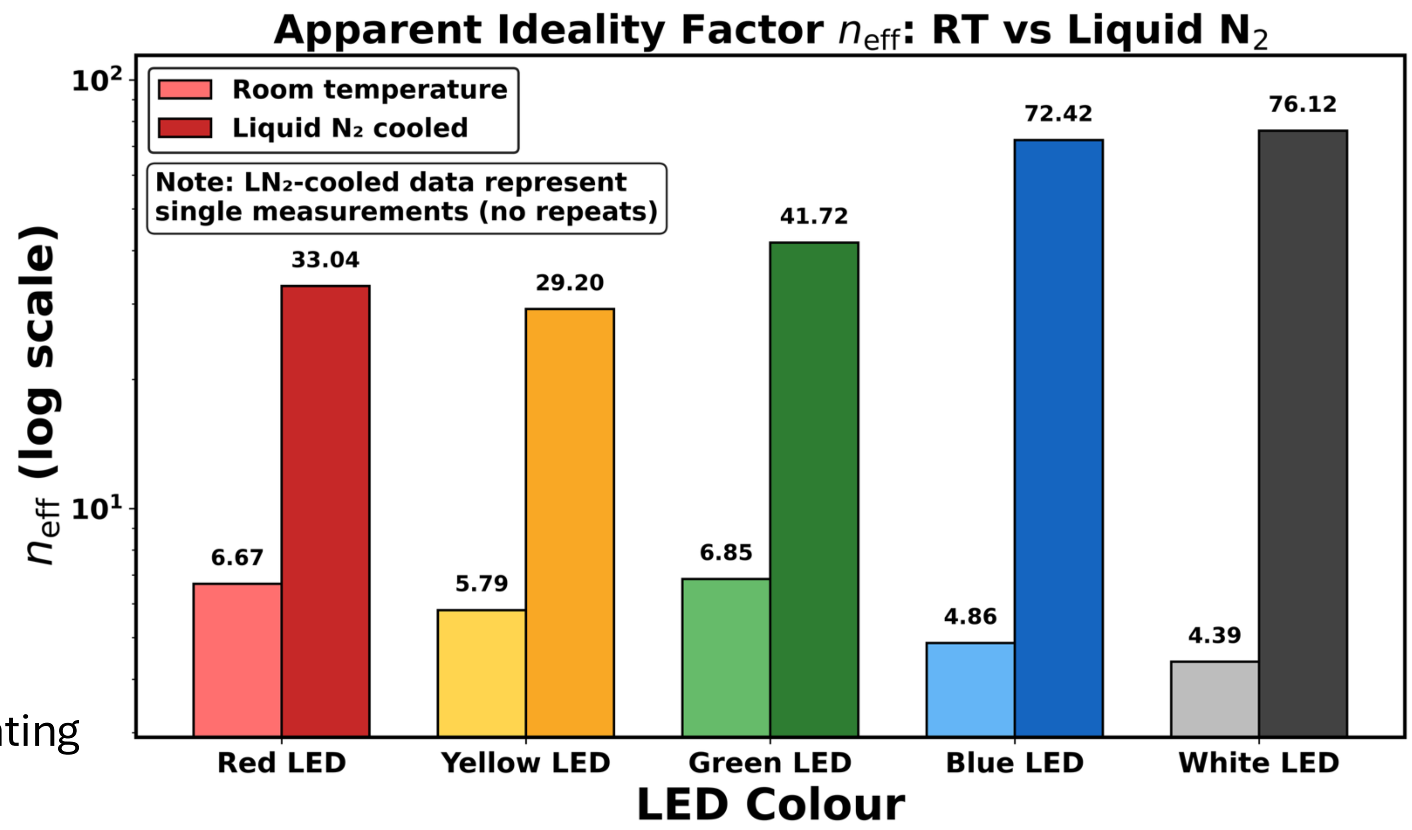
- n_{eff} was extracted using **Shockley diode equation** in post turn-on regions.

Room-temperature behaviour

- The extracted n_{eff} at RT are **significantly higher** than the ideal values (cf. $n = 1 - 2$) [4], reflecting **non-ideal transport mechanisms**, e.g., series resistance, carrier recombination, contact resistance, and inhomogeneous junctions in commercial LEDs.

Effect of LN_2 cooling

- Upon cooling to 77 K, n_{eff} increase further, indicating that **additional non-ideal effects dominate**.



Model applicability

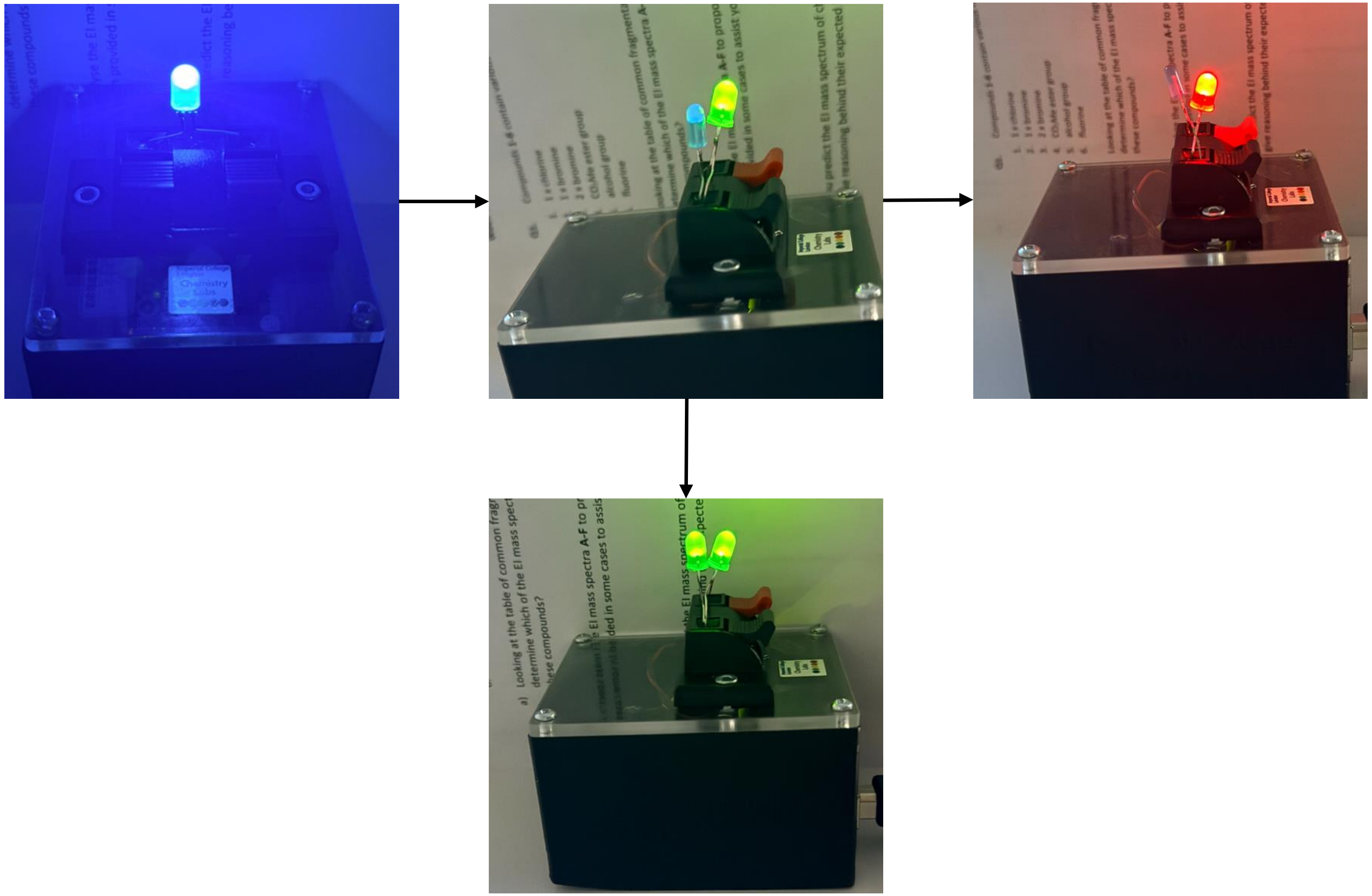
- At cryogenic temperatures, the assumptions underlying the **Shockley diode model** become less valid. Therefore, the extracted values should be interpreted as **apparent ideality factors**, useful for comparative trends rather than absolute material parameters.

[5]. S. M. Sze, Y. Li and K. K. Ng, Physics of Semiconductor Devices, 4th edn, Wiley, Hoboken, NJ, 2021.

Brightness distribution 1

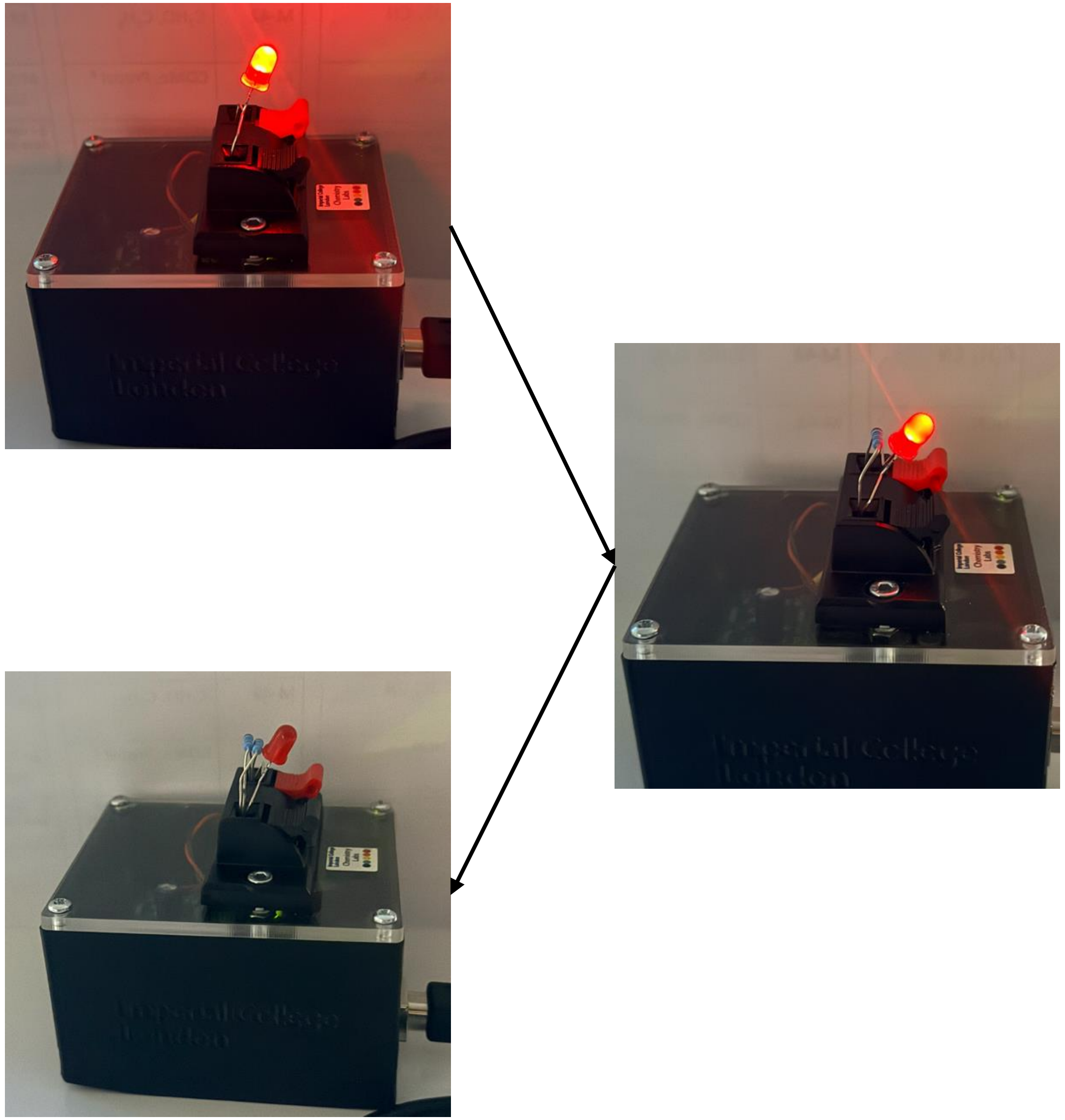
- When LEDs with **different turn-on voltages** are connected in parallel **without individual current-limiting**, the LED with the **lowest forward voltage** appears brighter.
- This results in an **asymmetric brightness distribution**, as current preferentially flows through the lowest–turn-on-voltage device.

- In contrast, **LEDs of the same type** (e.g. two green LEDs) exhibit **similar turn-on voltages** and therefore share current more evenly.
- As a result, LEDs with **matched electrical characteristics** display **comparable brightness** under identical bias conditions.



Brightness distribution 2

- For a **single LED**, adding a **series resistor** limits the forward current, resulting in a **progressive reduction in brightness**.
- This contrasts with the previous parallel-LED case, where **uncontrolled current sharing** led to asymmetric brightness.
- The observation confirms that **LED brightness is governed by current rather than voltage**.
- Series resistors therefore *can* provide **stable current control**, preventing dominance by low–turn-on-voltage LEDs and ensuring predictable emission.



Limitations & Future Work

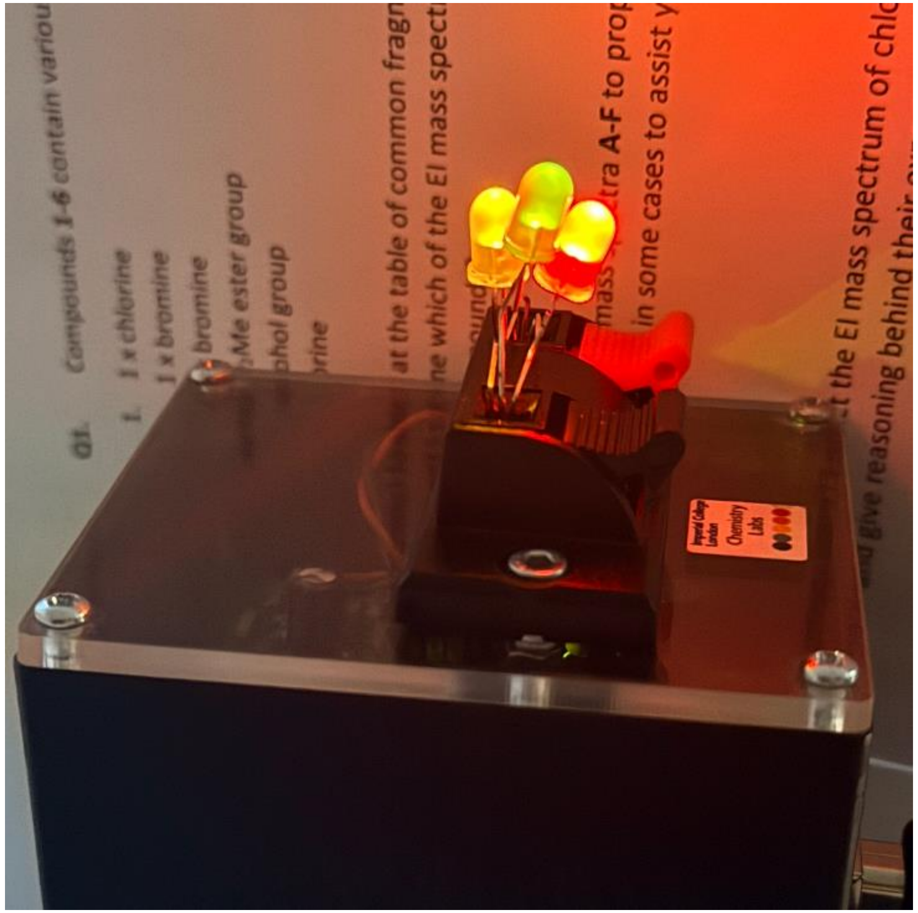
Experimental limitations

- The **bandgap energies of the LED materials could not be quantitatively determined** due to the absence of emission spectrum measurements.
- The **I–V characteristics at 77 K were measured only once for each LED**, owing to laboratory safety constraints and limited access to liquid nitrogen.
- At higher current levels, **Joule heating may increase the LED junction temperature**, introducing deviations from ideal I–V behaviour, particularly at room temperature.
- **Commercially packaged LEDs with unspecified semiconductor materials and doping levels** were used, limiting direct quantitative comparison with theoretical bandgap values and reported ideality factors in the literature.

Future work

- **Emission spectroscopy measurements** could be performed to **directly** determine the optical bandgap energies of the LED materials and enable quantitative comparison with turn-on voltages.
- **Repeated I–V measurements at 77 K** could be carried out under controlled cryogenic access to improve statistical reliability and allow uncertainty analysis at low temperature.
- **Active thermal management or pulsed-current measurements** could be implemented to minimise Joule heating effects and better isolate intrinsic diode behaviour at higher currents.
- **Use of LEDs with known material composition and doping specifications**, or custom-fabricated devices, would allow more rigorous comparison with Shockley theory and reported ideality factors.

Fun fact



Traffic lights can be made in this way!

SI 1.1: Serial communication principle

Serial data transmission

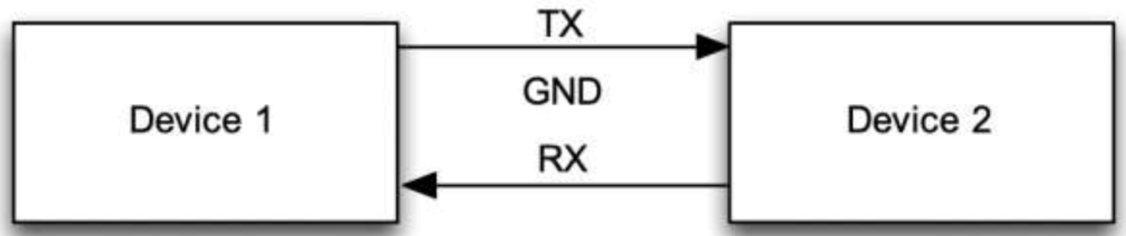
- Data are sent one bit at a time over a **single** wire (TX), rather than in parallel.
- Communication uses **separate TX and RX lines** with a shared ground (GND) reference.
- In this experiment, serial communication is used to **send control commands** to the Arduino and **stream measurement data** back to the computer.

Data framing (8N1)

- Each byte is transmitted as:
  Start bit (0) → 8 data bits → Stop bit (1)
- No parity is used in this configuration

UART communication on the Arduino

- The Arduino microcontroller uses **UART (Universal Asynchronous Receiver–Transmitter)** for serial communication
- UART defines the **timing and framing** of serial data transmission (baud rate, start/stop bits).
- In this experiment, the UART operates at a **baud rate of 9600** with **8N1 framing**, matching the serial configuration used by the host computer.



Schematic of protocol., figure taken from experimental manual

In this experiment, RS232-style serial conventions are followed at the protocol level.

Classical RS232 Binary signalling

- +3 to +15 V → **logic 0 (HIGH)**
- –3 to –15 V → **logic 1 (LOW)**
- -3 to +3 V → **dead zone (to tolerate noise)**
- Line remains LOW (1) when in **idle state**

On the Arduino, UART logic levels are **TTL-level voltages** [6]:

- LOW ≈ 0 V
- HIGH ≈ 5 V
- The line is **HIGH when idle**

[6]: Arduino Programming Language Document Serial Functions, https://docs.arduino.cc/language-reference/en/functions/communication/serial/, (accessed 28 January 2026).

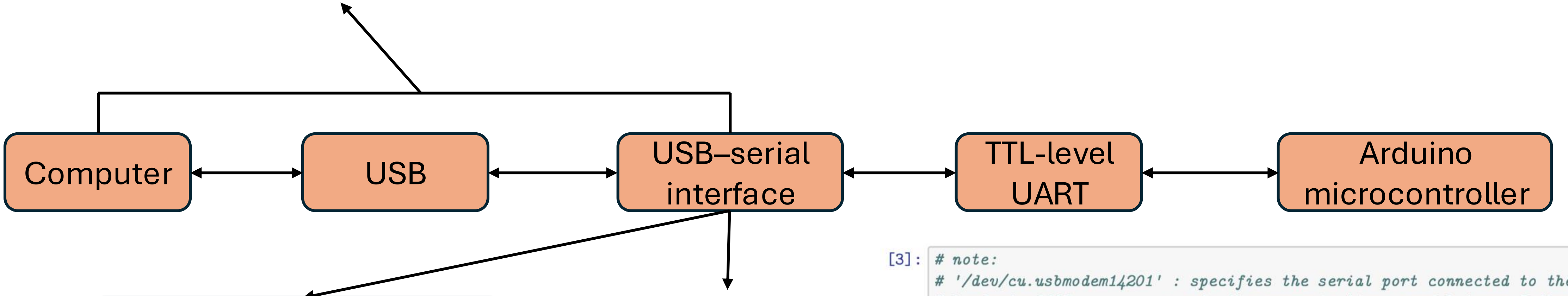
SI 1.2: Arduino communication

Communication pathway

- Arduino is physically connected to the computer via **USB**.
- An on-board **USB–serial interface** presents the device as a **virtual serial (COM) port**.
- Serial data on the Arduino uses **TTL-level UART**

Application-level serial protocol

- **ASCII-encoded** commands and data are sent between the computer and Arduino
- Command format: **<An>**
  - A: command identifier
  - n: numeric parameter (int or float)
- Arduino responses terminated by **\r\n**
- Host software reads using **\n** for reliable parsing



Note: RS232-style conventions (baud rate, framing, ASCII data) are used at the **protocol level**.
Electrical signalling on the Arduino uses **TTL-level UART**, enabled via an on-board USB–serial interface.

Serial configuration

- Baud rate: **9600**
- Data framing: **8N1**
- Asynchronous communication

```
[3]: # note:
     # '/dev/cu.usbmodem14201' : specifies the serial port connected to the device
     # baudrate=9600           : sets the communication speed between Python and the
      ↪device
     # timeout=2 s             : limits how long the program waits for a response

     ser = serial.Serial('/dev/cu.usbmodem14201', baudrate=9600, timeout=2)
```

Example host-side serial initialisation (Python / pySerial)

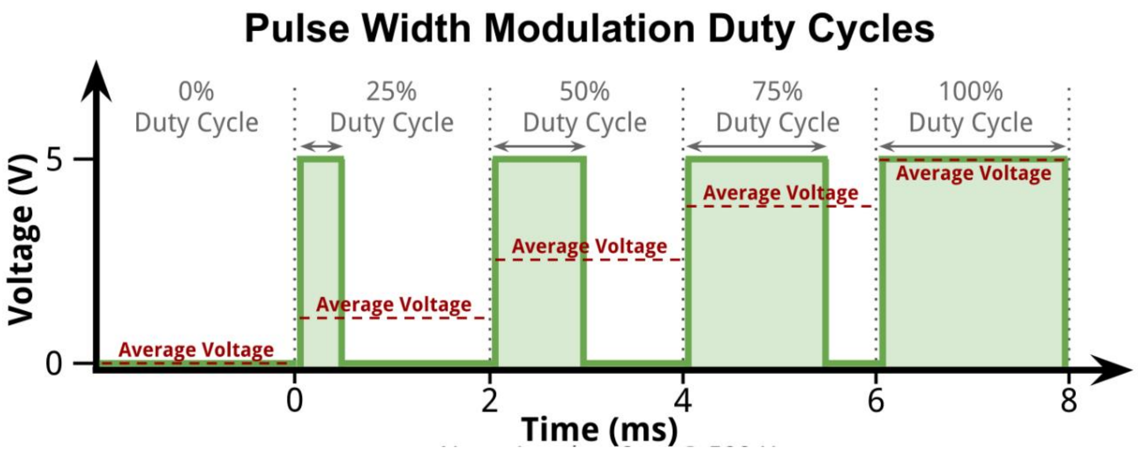
SI 2: Analogue output & PWM

- Arduino pins **D3 and D11 are digital outputs** (0 V or 5 V only)
- **Pulse width modulation (PWM)** approximates analogue voltage by rapidly switching between **high** and **low** states

- The **average output voltage** depends on the **duty cycle** D of PWM:

$$V_{out} = D \times 5\,V$$

- Varying D allows **continuous control** of the effective output voltage.



Example figure of PWM working principle, taken from [6]

Effective DC voltage generation

- Measurement timescale ≫ PWM period
- RC filter suppresses high-frequency ripple
- Effective DC voltage enables accurate I–V sweeps

- The PWM signal operates at **~64 kHz** (period ≈ **15.6 μs**), much faster than the measurement timescale.
- An **RC low-pass filter** (R1, R2, C1; see SI 3) smooths the PWM waveform into a **stable DC voltage**.
- The resulting **voltage ripple** is **< 1%**, enabling reliable I–V measurements.

[6]. Robotic Controls, Pulse Width Modulation with analogWrite, https://robotic-controls.com/book/export/html/57/(accessed 26 January 2026).

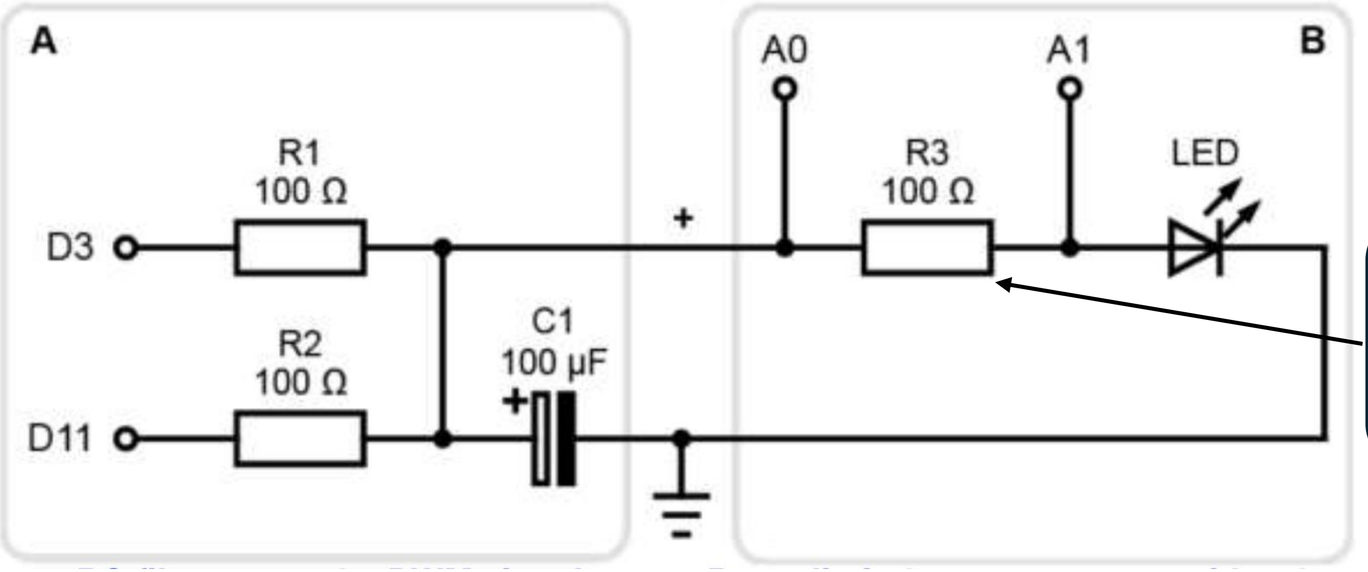
SI 3: Circuit structure and component roles

PWM generation (D3, D11)

- Arduino digital pins output a high-frequency PWM signal (0–5 V), which is used as a controllable voltage source

Differential voltage measurement (A0, A1)

- Measuring across R3 reduces sensitivity to supply fluctuations and improves current measurement accuracy compared to single-ended sensing.



*The resistance is **large enough** to prevent excessive current and protect the LED, yet **small enough** to generate a measurable voltage drop within the ADC resolution.*

Figure taken from lab manual

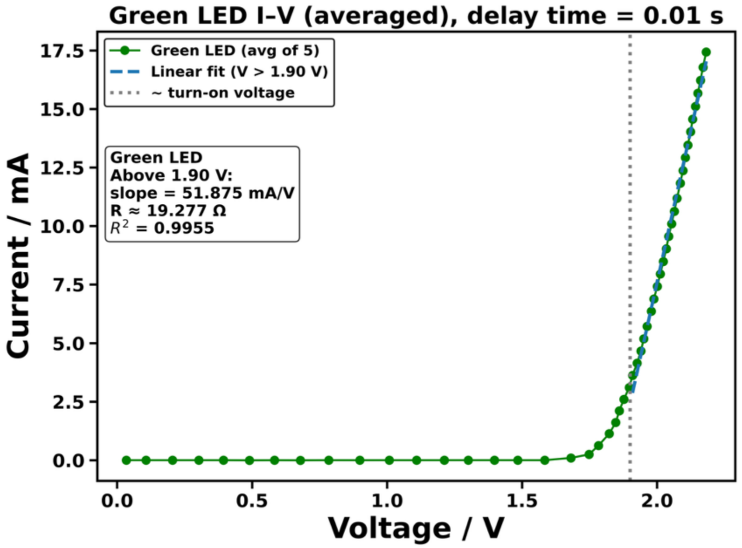
RC low-pass filter (R1, R2, C1)

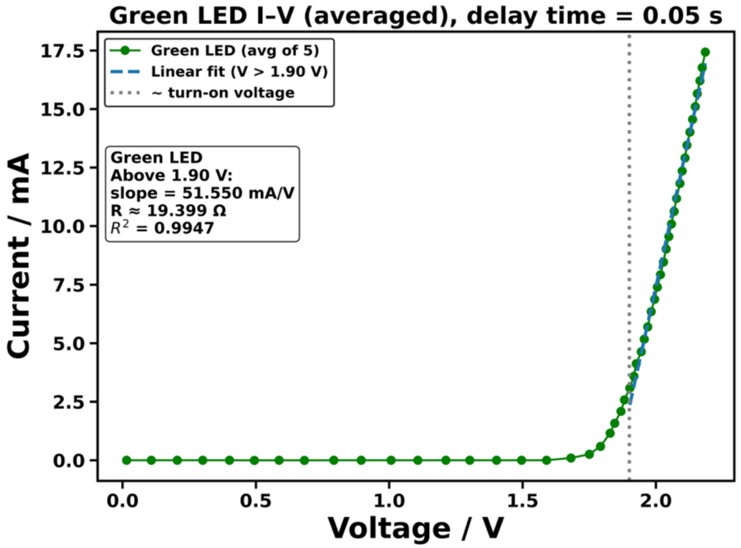
- The RC network smooths the PWM waveform into a **stable DC voltage**, removing high-frequency components before the signal reaches the measurement circuit.

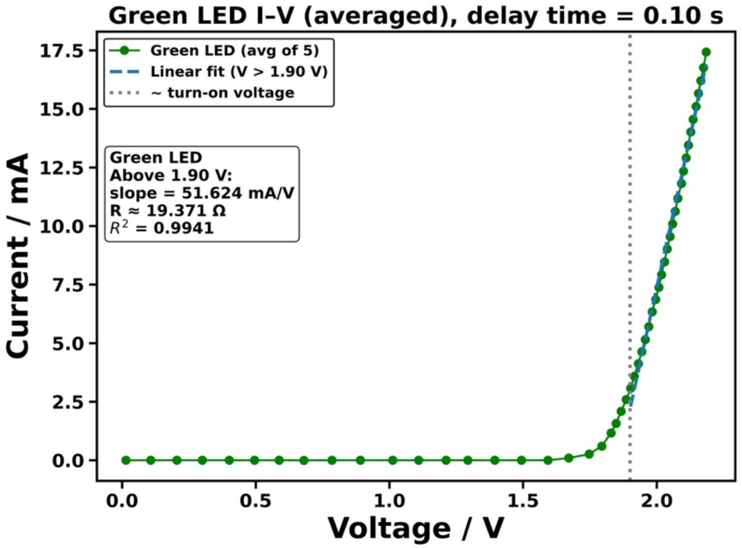
Sensing resistor (R3)

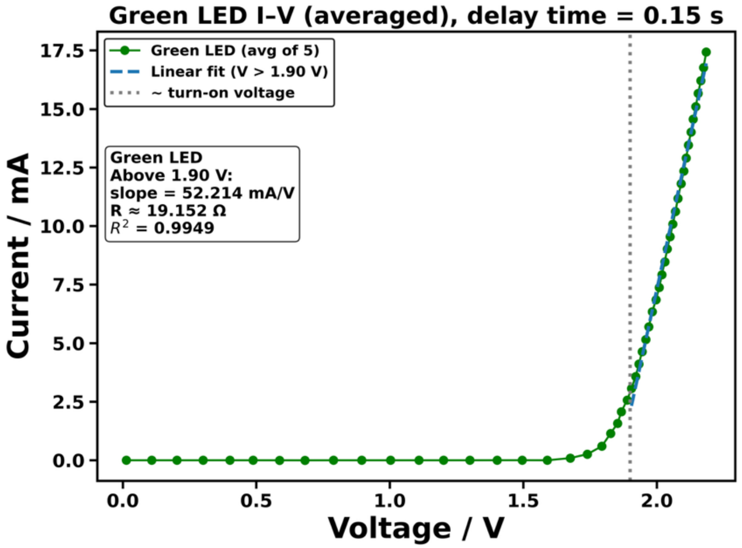
- **limits the LED current** to a safe range
- provides a **known resistance** for calculating current via $I = (V_{A0} - V_{A1})/R_3$

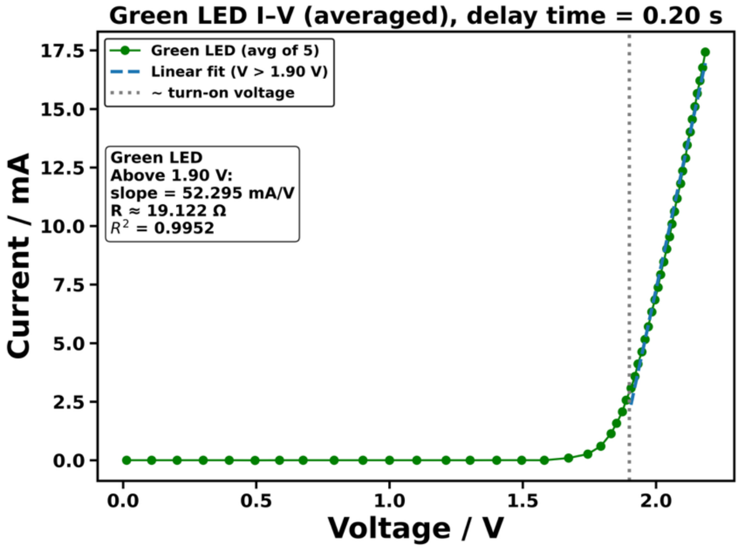
SI 4: Delay Time Figures

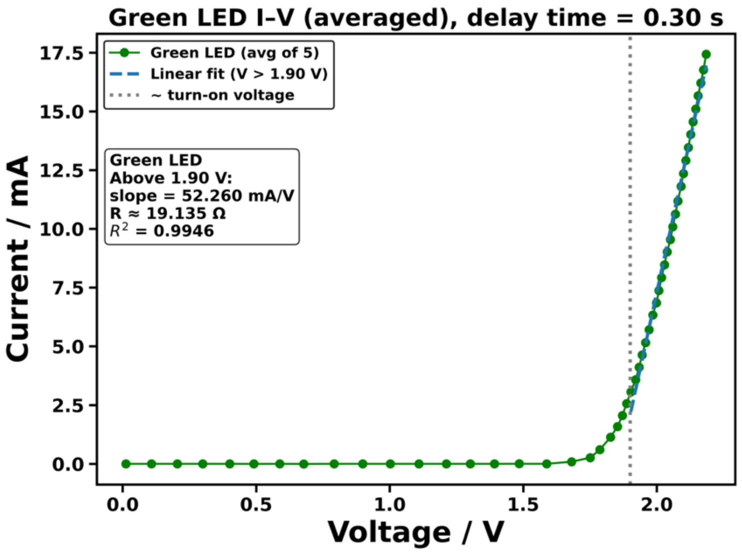




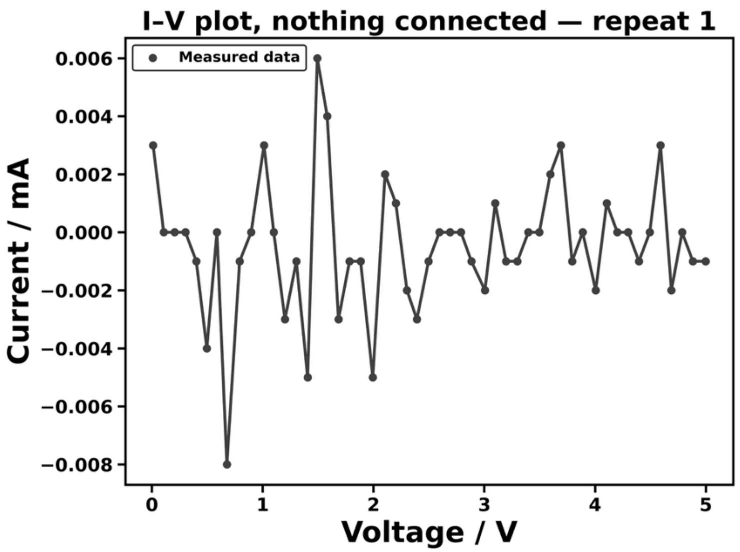


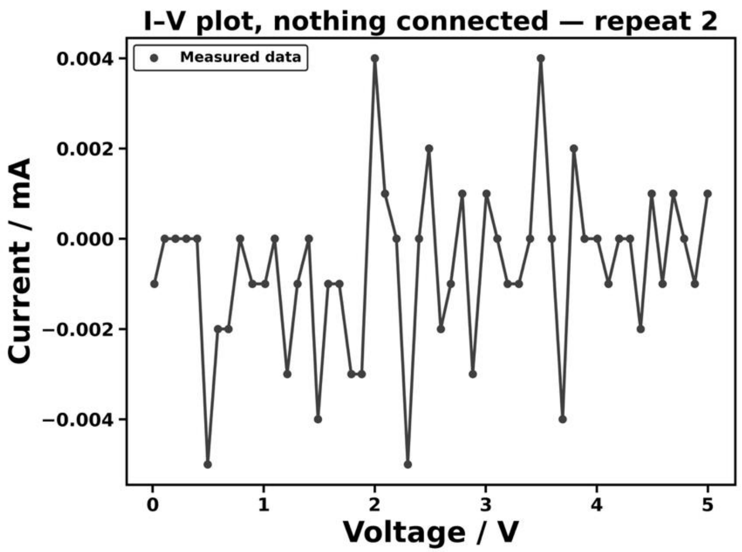


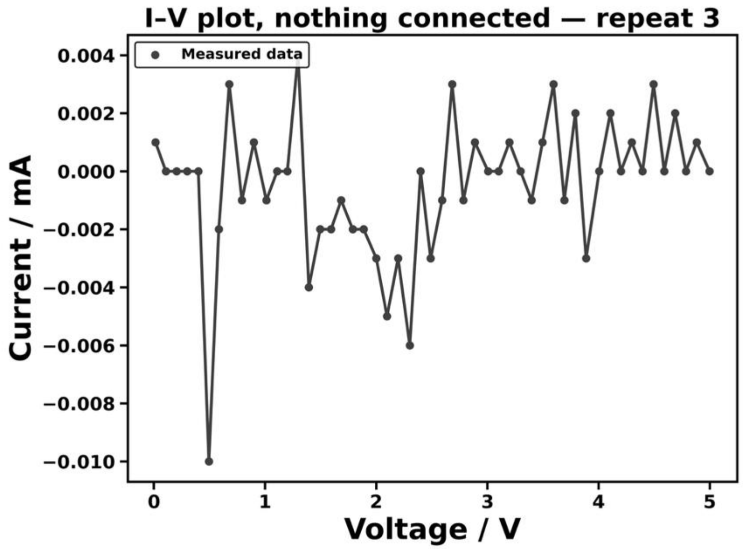


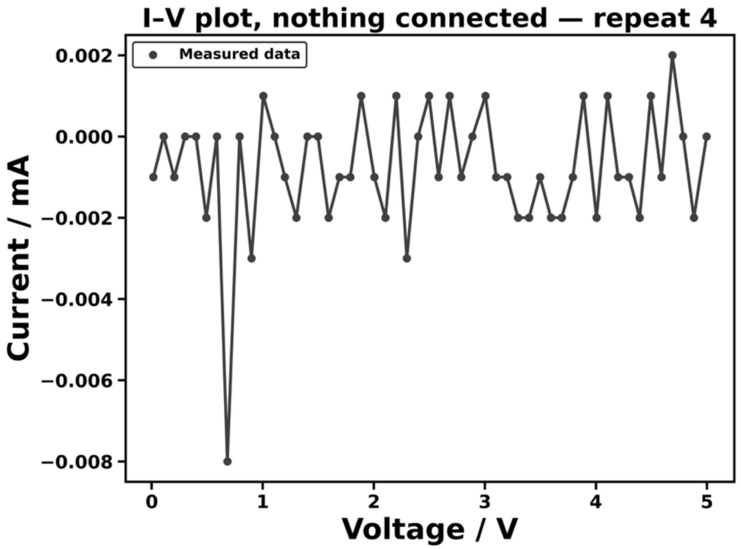


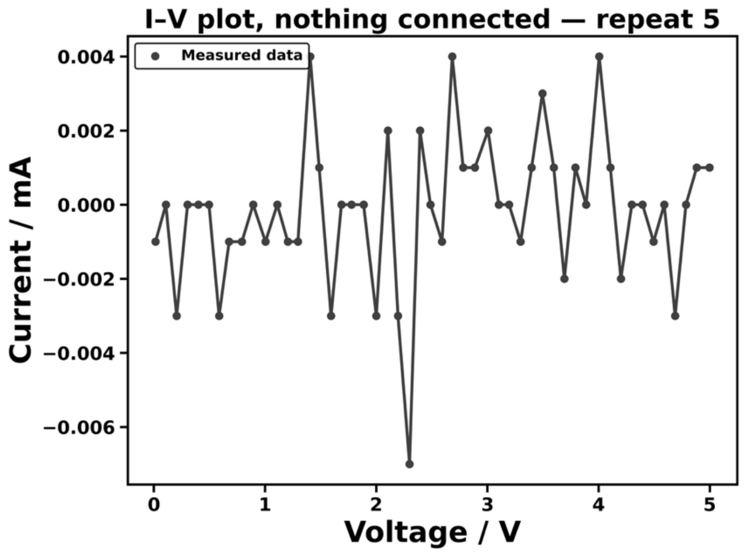
SI 5: open circuit figures



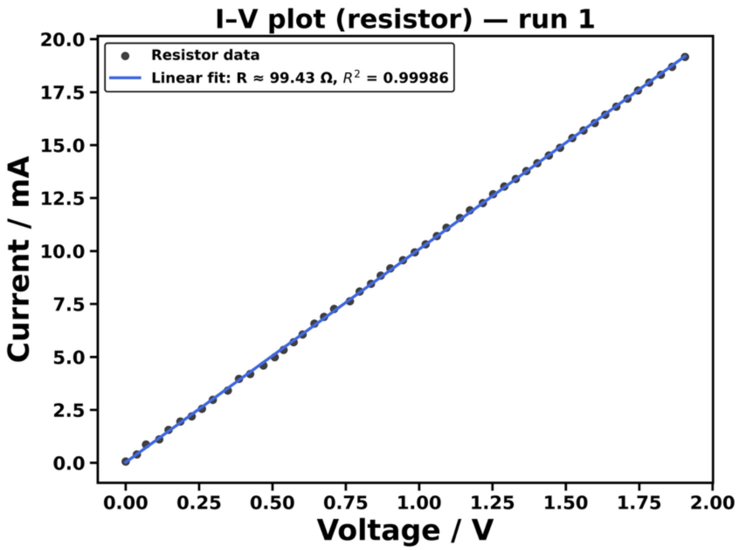


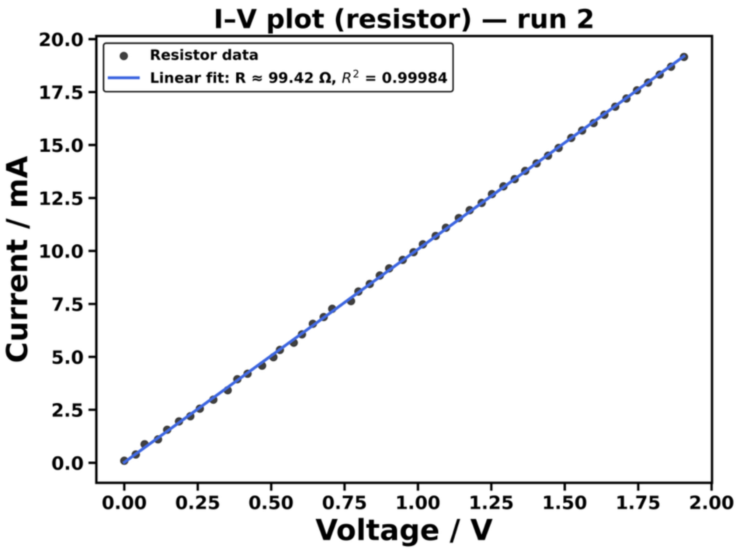


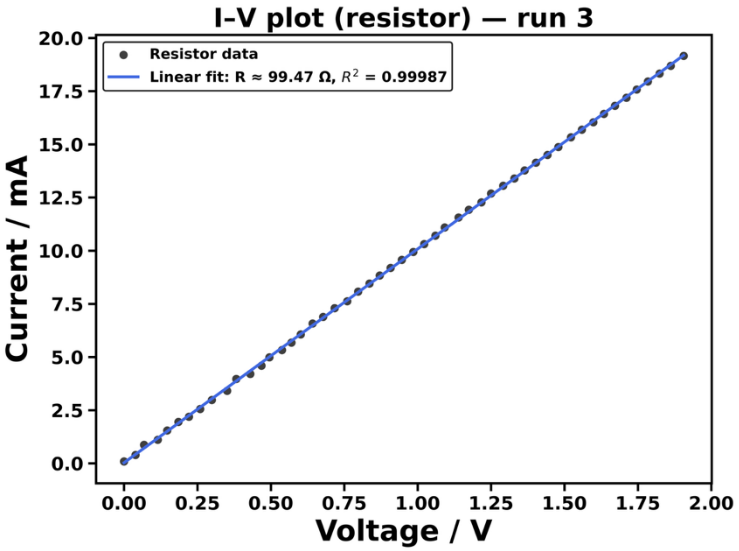


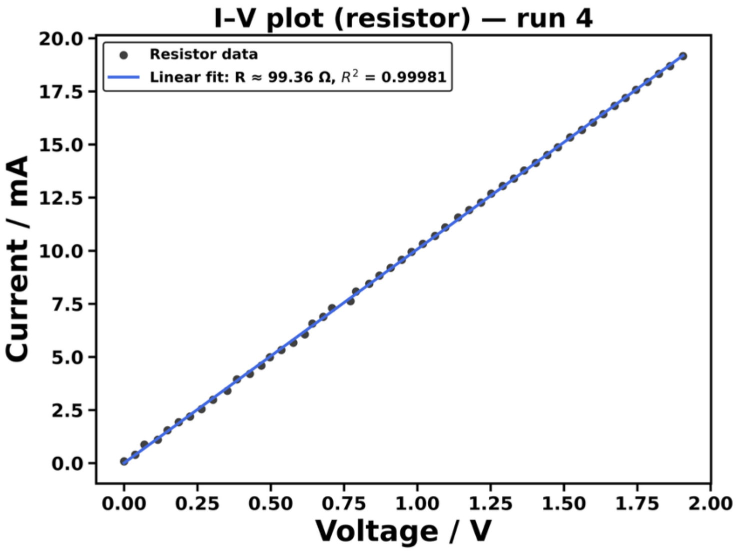


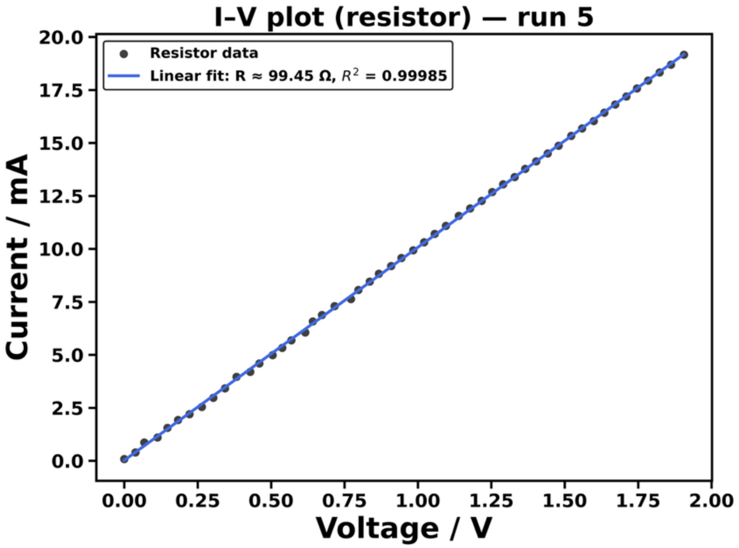
SI 6.1: resistor figures



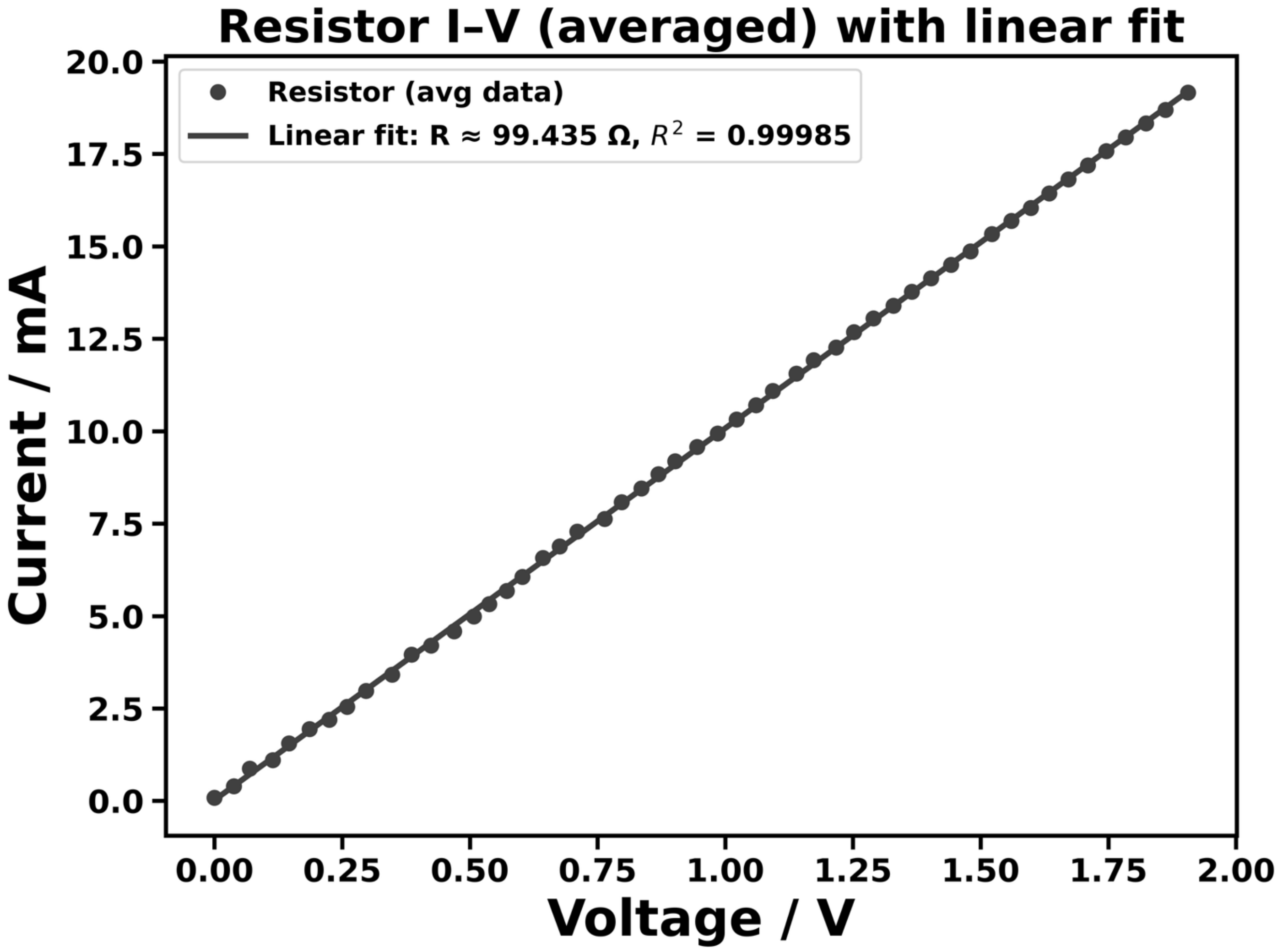




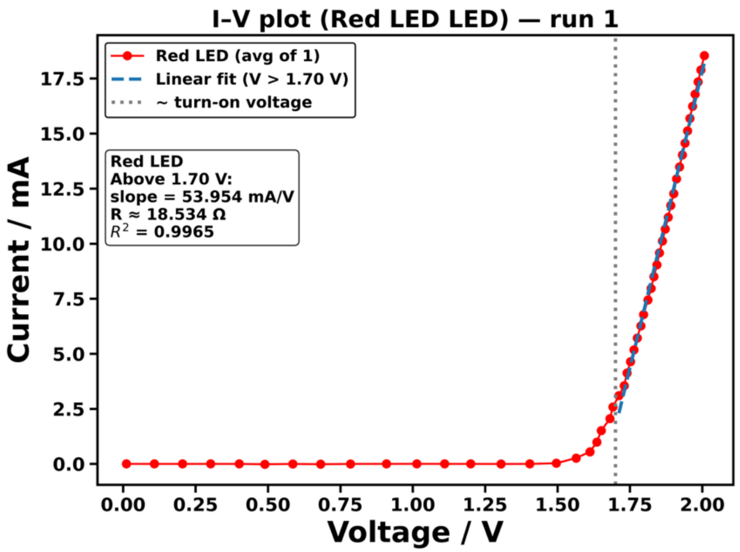


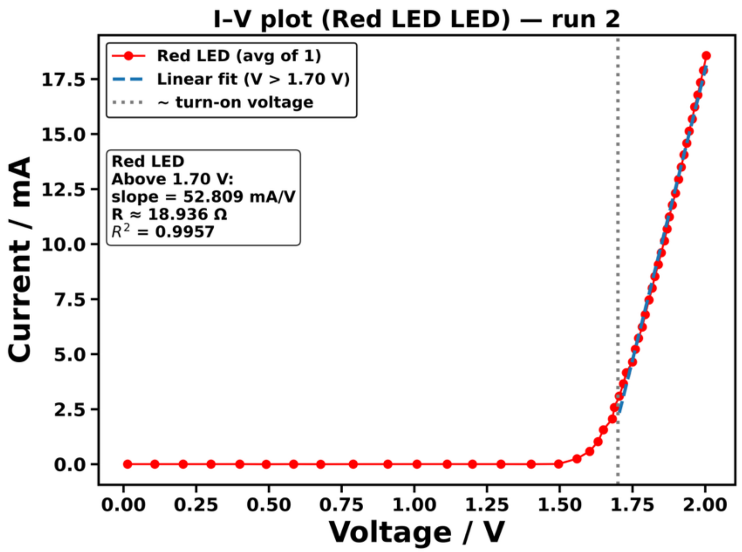


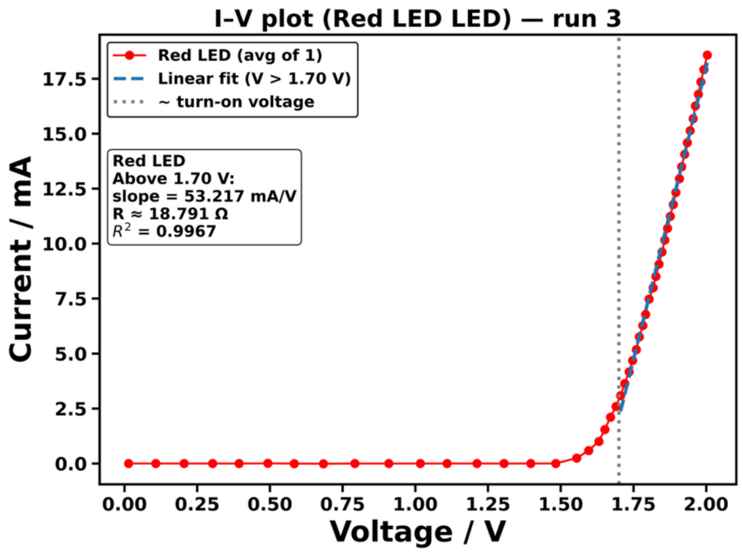
SI 6.2: linear fit of resistor

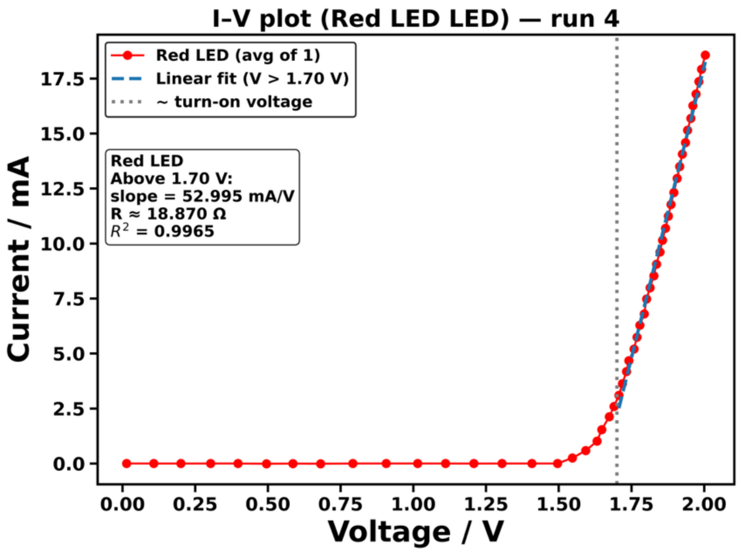


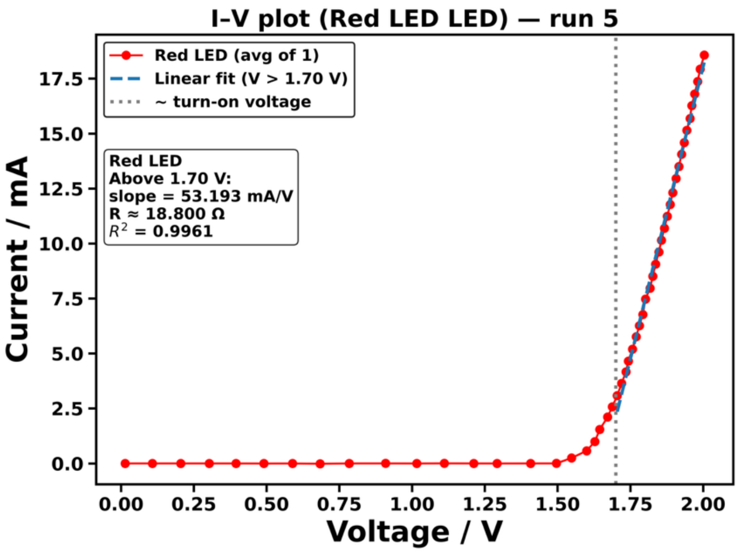
SI 7.1: red LED


I-V plot (Red LED LED) — run 1
Red LED (avg of 1)
Linear fit (V > 1.70 V)
~ turn-on voltage
Red LED
Above 1.70 V:
slope = 53.954 mA/V
R ≈ 18.534 Ω
R^2 = 0.9965
Current / mA
Voltage / V
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

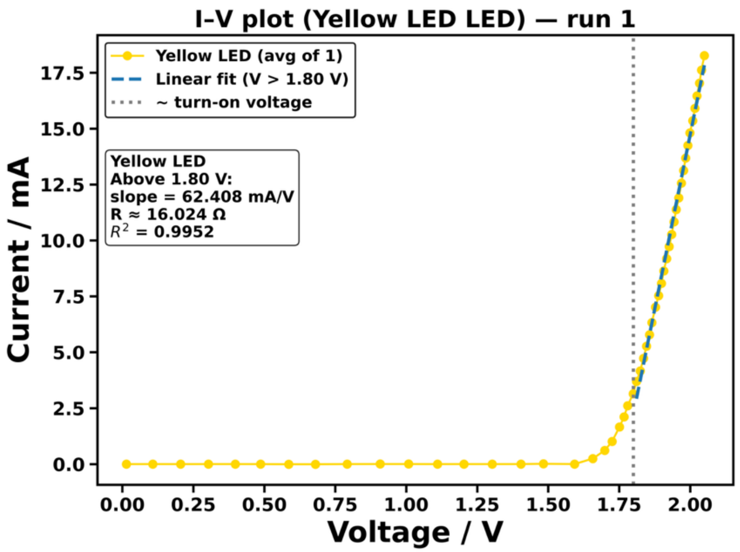

I-V plot (Red LED LED) — run 2
Red LED (avg of 1)
Linear fit (V > 1.70 V)
~ turn-on voltage
Red LED
Above 1.70 V:
slope = 52.809 mA/V
R ≈ 18.936 Ω
R^2 = 0.9957
Current / mA
Voltage / V
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

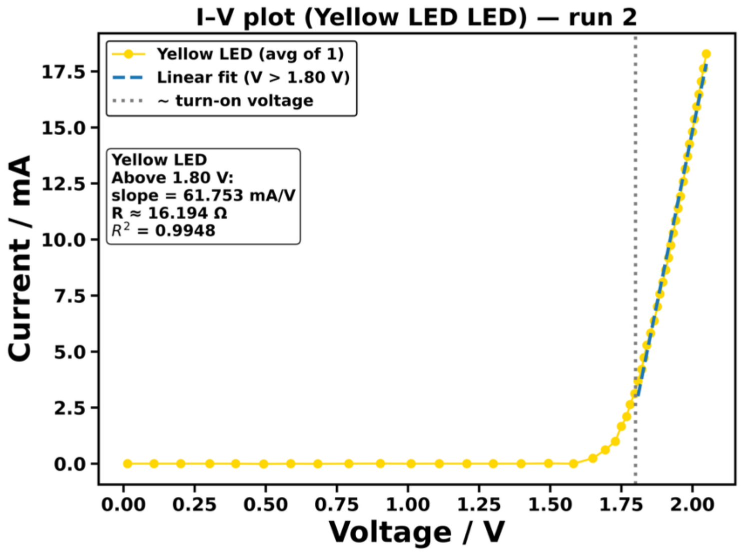

I-V plot (Red LED LED) — run 3
Red LED (avg of 1)
Linear fit (V > 1.70 V)
~ turn-on voltage
Red LED
Above 1.70 V:
slope = 53.217 mA/V
R ≈ 18.791 Ω
R^2 = 0.9967
Current / mA
Voltage / V
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

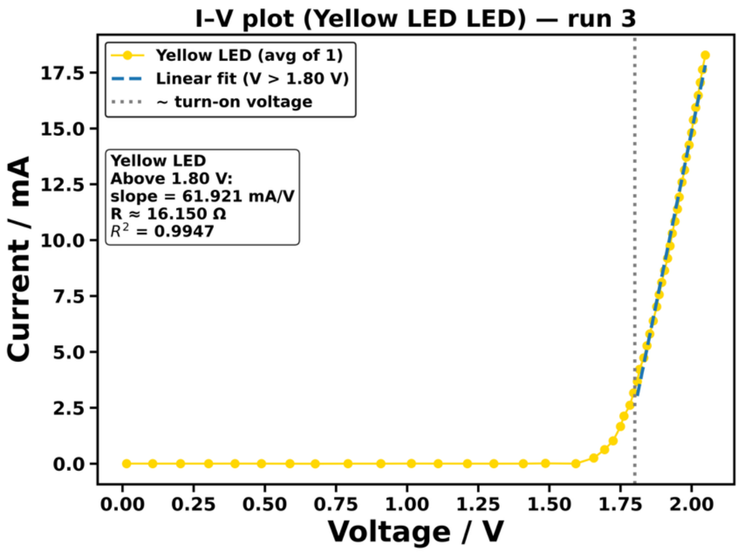

I-V plot (Red LED LED) — run 4
Red LED (avg of 1)
Linear fit (V > 1.70 V)
~ turn-on voltage
Red LED
Above 1.70 V:
slope = 52.995 mA/V
R ≈ 18.870 Ω
R^2 = 0.9965
Current / mA
Voltage / V
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

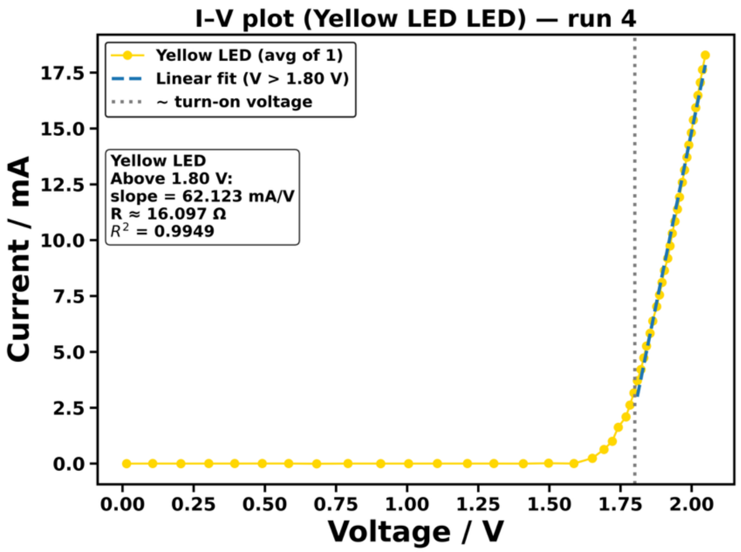

I-V plot (Red LED LED) — run 5
Red LED (avg of 1)
Linear fit (V > 1.70 V)
~ turn-on voltage
Red LED
Above 1.70 V:
slope = 53.193 mA/V
R ≈ 18.800 Ω
R^2 = 0.9961
Current / mA
Voltage / V
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

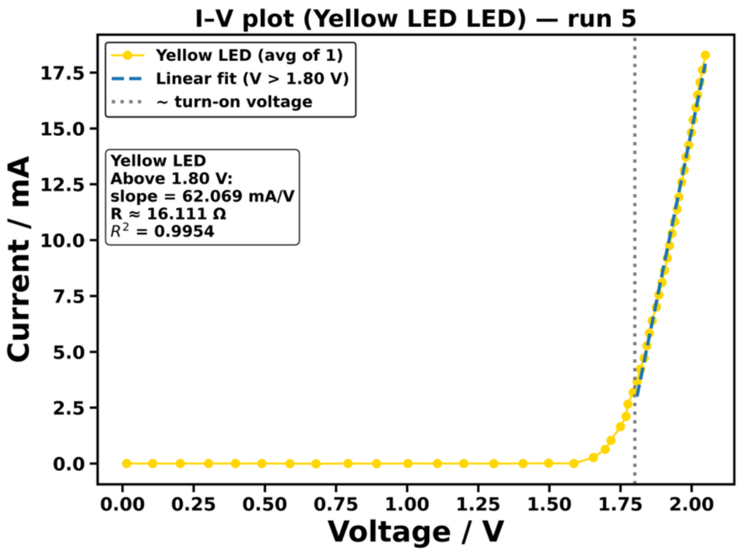
SI 7.2: yellow LED



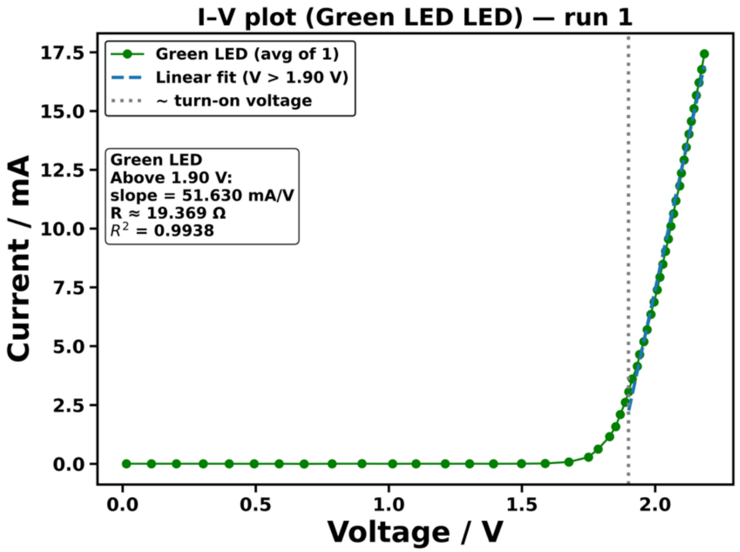


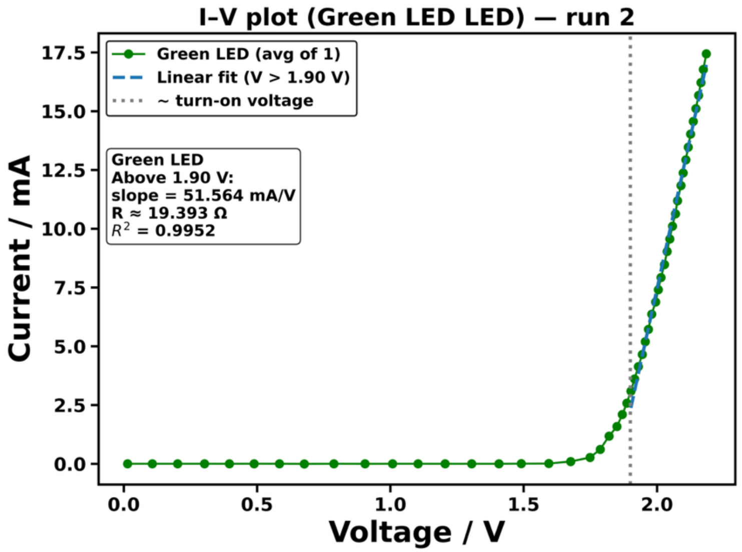


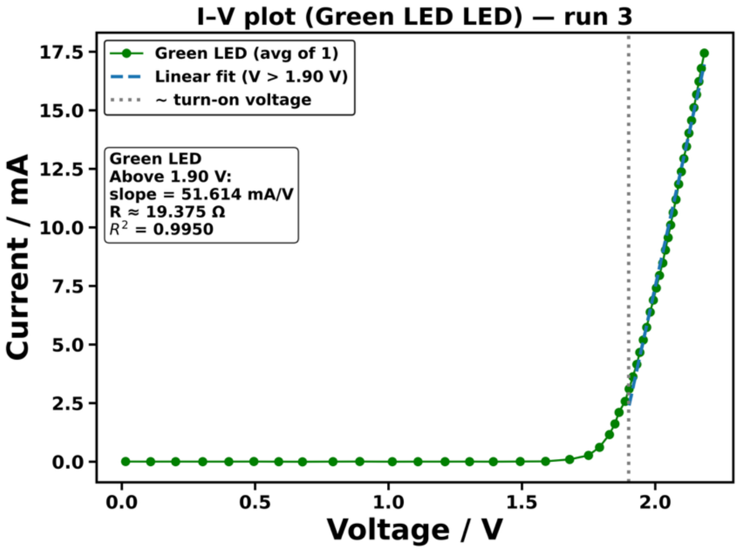


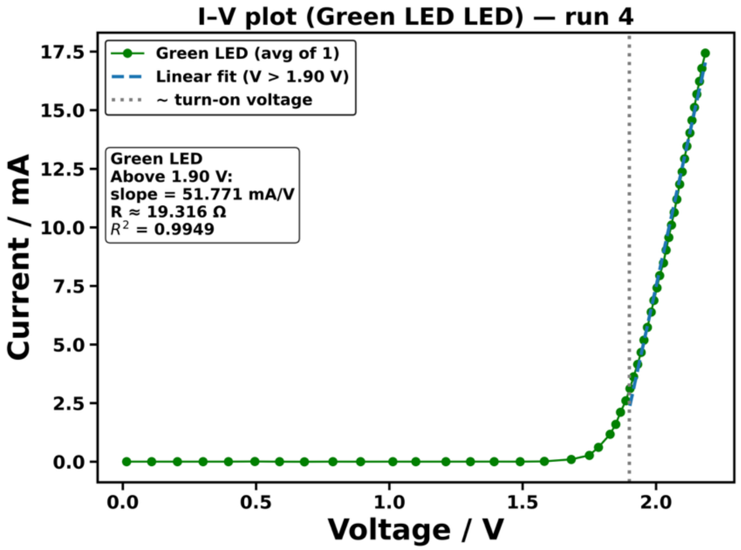


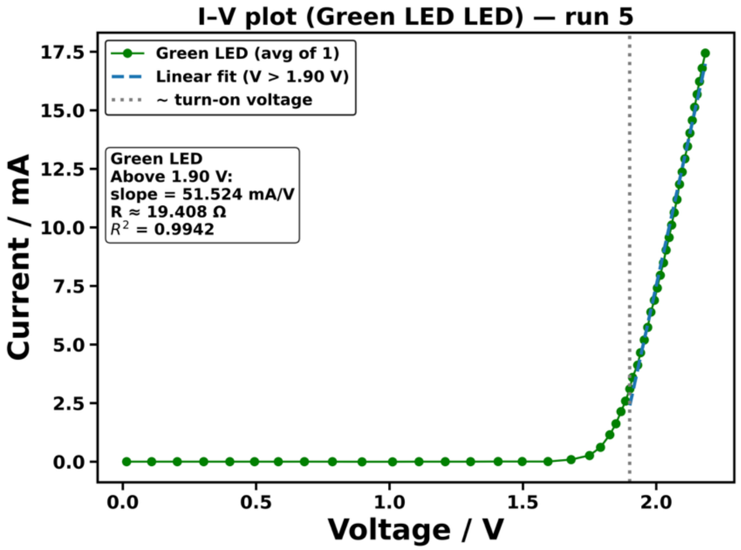
SI 7.3: green LED



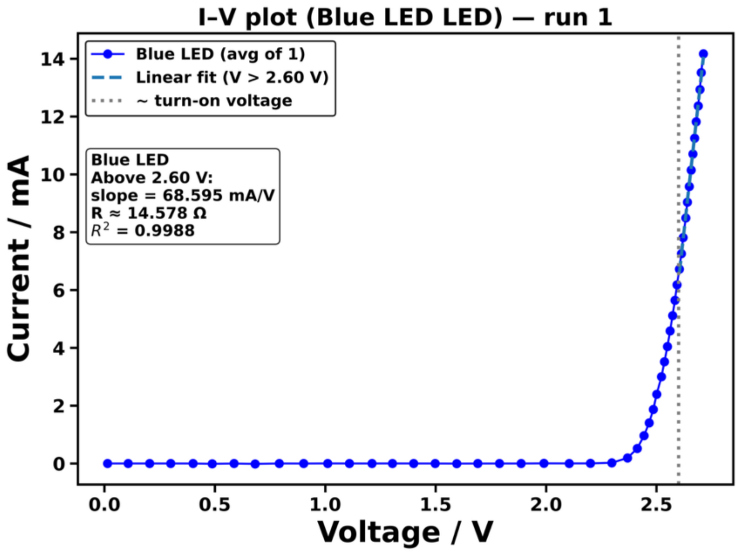


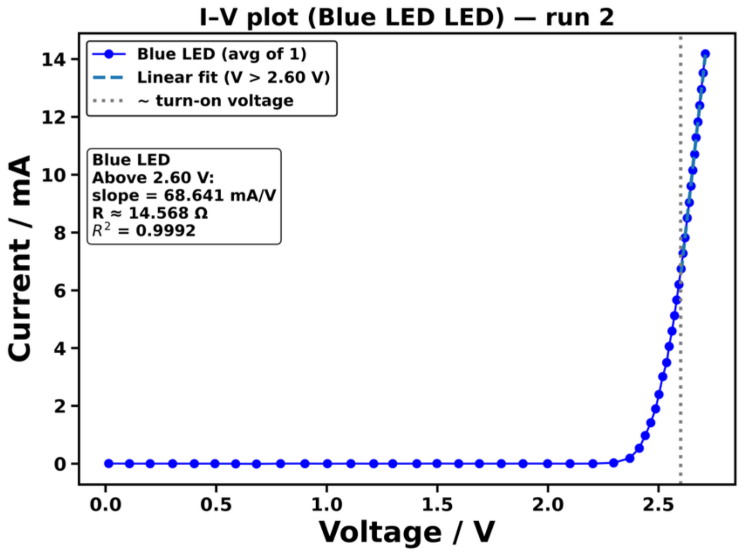


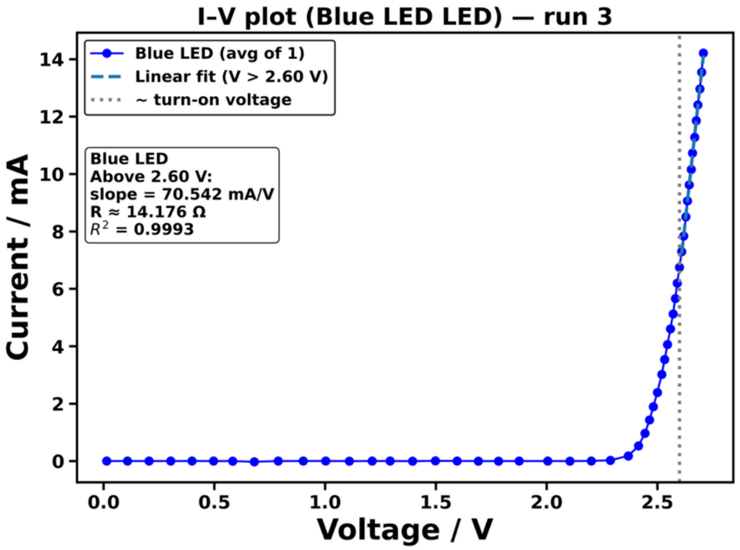


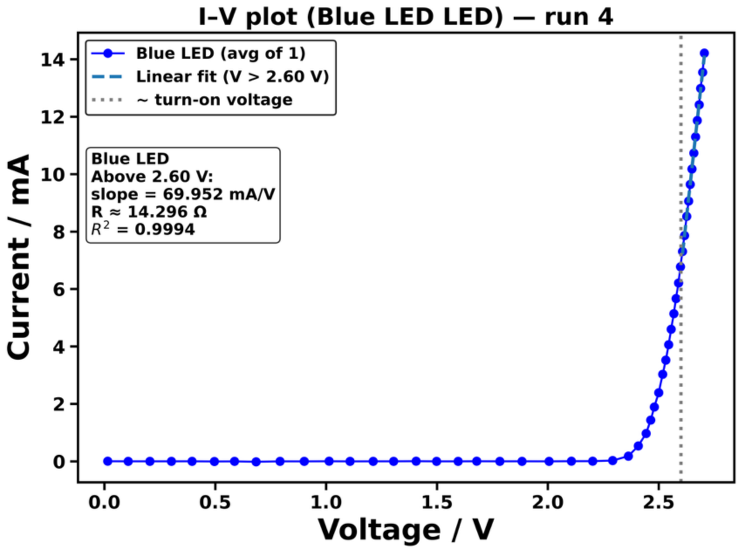


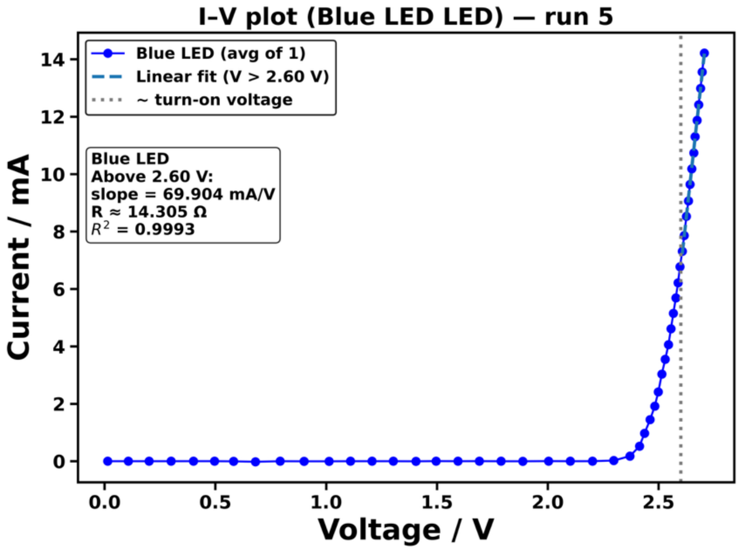
SI 7.4: blue LED



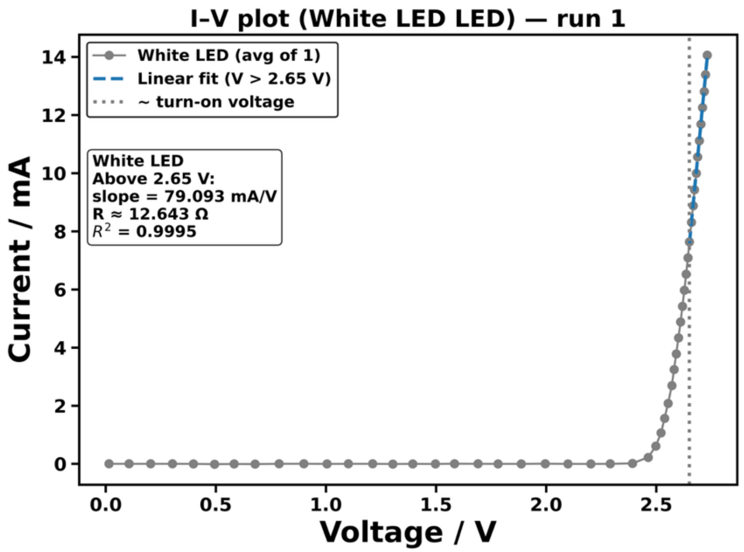


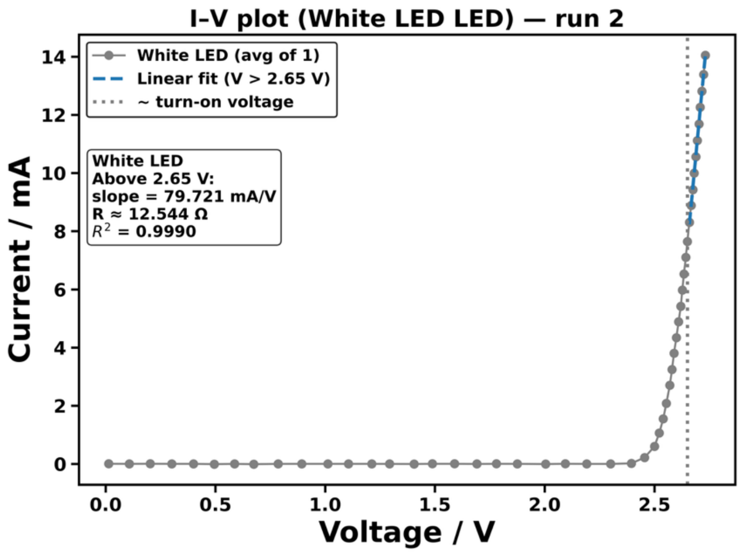


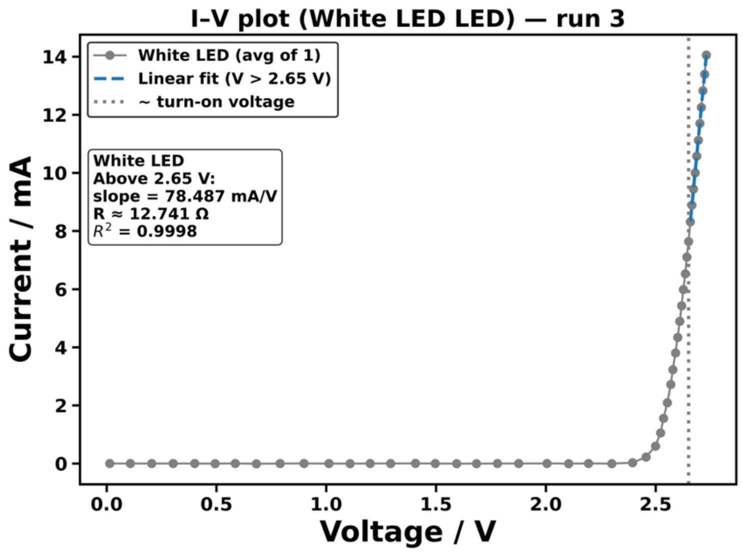


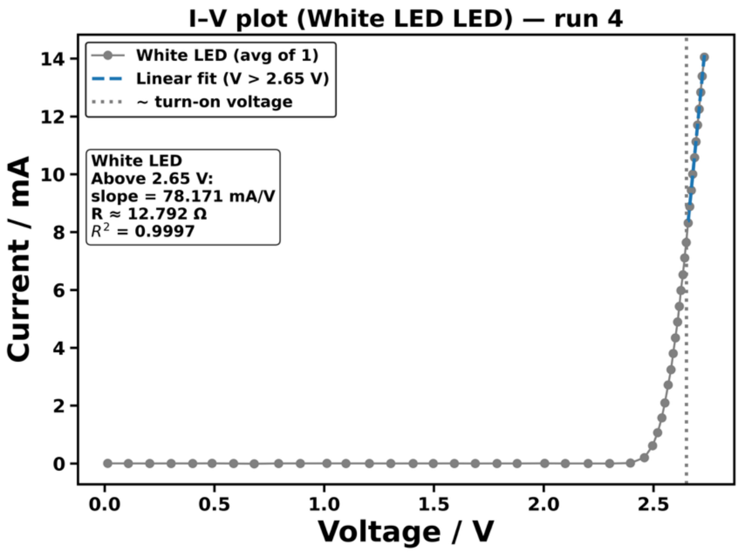


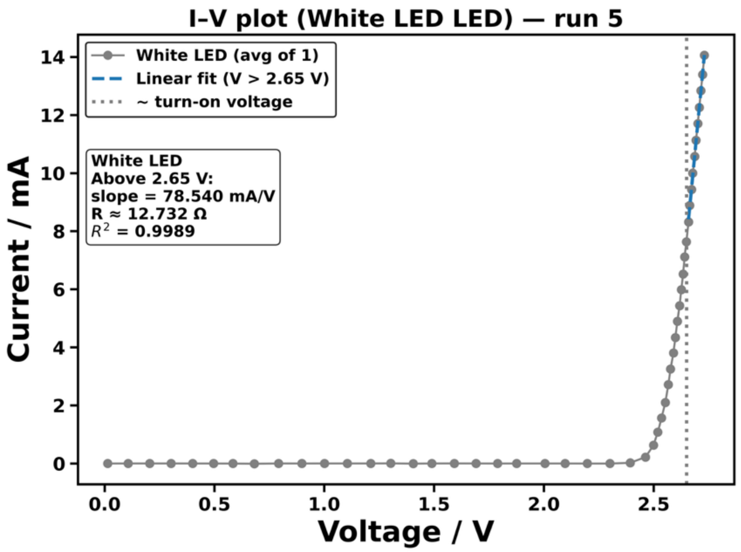
SI 7.5: white LED











SI 8: $\log(I) - V$ relation justification

Shockley diode equation: $I_D = I_S\left(e^{\frac{V_D}{nV_T}} - 1\right)$, V_D = voltage applied,

where $V_T = \frac{kT}{q}$; $V_T = 2.49\times10^{-2}\,V$ at r.t., $V_T = 6.64\times10^{-3}\,V$ at 77K. $\}$ I_S: saturation current

For moderate forward bias V_D, $e^{\frac{V_D}{nV_T}} >> 1$,

therefore $I_D \sim I_S e^{\frac{V_D}{nV_T}}$.

Take log of both sides: $\log I_D = \log\left(I_S e^{\frac{V_D}{nV_T}}\right)$

$$= \log I_S + \log e^{\frac{V_D}{nV_T}}$$

$$= \log I_S + \frac{V_D}{nV_T}\log e$$

$$= \log I_S + \frac{V_D}{nV_T \ln 10}.$$

V_T, $\ln 10$ = constant; n = constant for each material.

$$\Rightarrow \log I_D = \frac{1}{nV_T \ln 10} V_D + \log I_S$$

$$y = \underline{m}\; x + c$$

$\hookrightarrow$ slope $= \frac{1}{nV_T \ln 10} \Rightarrow n = \frac{1}{\text{slope} \cdot V_T \ln 10}$

SI 9: Reflection on the usage of automation in chemistry experiments

Take-home reflection:
Automation is most powerful when used to **systematically explore known parameter spaces**, but it complements rather than replaces human intuition during early experimental discovery.

Pros

Efficiency and scale
- High experimental throughput
- Continuous operation with minimal supervision
- Enables systematic and parallel measurements
- Accelerates optimisation and data collection

Reproducibility and precision
- Reduces human handling and timing errors
- Ensures fixed, repeatable experimental sequences

Safety and data integrity
- Minimises direct exposure to hazardous conditions
- Direct digital data capture reduces transcription errors

Cons

Reduced flexibility during discovery
Automation works best when:
- Protocols are stable
- Parameter space is known
- Objectives are defined

<u>But early-stage chemistry often needs</u>:
- Trial-and-error experimentation
- Rapid procedural changes
- Evolving hypotheses

Initial development and maintenance cost
Automatic systems requires
- Upfront hardware and software investment
- Ongoing debugging, calibration, and maintenance

In this experiment, **significant time was spent repeatedly in developing and adjusting the control and data-analysis scripts**, highlighting the development overhead of automation.